



Modelling Financial Markets

Chapter 1: Stylised Facts and Returns



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Acronyms used in this material (1/2)

- ▣ **ACF**: AutoCorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARCH**: AutoRegressive Conditional Heteroskedasticity
- ▣ **ARCH-LM**: ARCH Lagrange Multiplier (test)
- ▣ **BET-TR**: BET Total Return (index)
- ▣ **BNR**: Banca Națională a României — *the National Bank of Romania*
- ▣ **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- ▣ **CAGR**: Compound Annual Growth Rate
- ▣ **CI**: Confidence Interval
- ▣ **CLT**: Central Limit Theorem
- ▣ **CRIX**: CRypto IndeX
- ▣ **DAX**: Deutscher Aktienindex — *the German stock index*
- ▣ **EGARCH**: Exponential GARCH
- ▣ **ETF**: Exchange-Traded Fund
- ▣ **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- ▣ **GEO**: Government Emergency Ordinance
- ▣ **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HAR-RV**: Heterogeneous AutoRegressive model of Realised Volatility
- ▣ **JB**: Jarque–Bera (test)
- ▣ **LB**: Ljung–Box (test)
- ▣ **LM**: Lagrange Multiplier
- ▣ **MDD**: Maximum DrawDown
- ▣ **OHLC**: Open, High, Low, Close
- ▣ **PDV**: Path-Dependent Volatility
- ▣ **QQ**: Quantile–Quantile (plot)
- ▣ **RMSE**: Root Mean Squared Error
- ▣ **SE**: Standard Error
- ▣ **TSPL**: Time-Shifted Power Law (kernel)
- ▣ **US**: United States
- ▣ **VIX**: CBOE Volatility Index
- ▣ **VIX9D**: Cboe 9-Day Volatility Index

Chapter Route and the Definition of a Stylised Fact

Route of this chapter

- Part I: returns, distribution and tails, dependence and its mechanisms
- Part II
 - ▶ persistence and the limits of inference, market exceptions
 - ▶ volatility from open, high, low and close prices
 - ▶ performance, a case study
 - ▶ one pass of the AI-assisted research loop
- Technical detail sits in the [Appendix](#) at the end

Definition

- A **stylised fact** is a qualitative statistical regularity of **returns**
 - ▶ seen across assets, markets and periods [Cont \(2001\)](#)
 - It remains a hypothesis [Ratliff-Crain, Van Oort, Koehler & Tivnan \(2025\)](#)
 - ▶ to be re-tested on each market and frequency
 - Key message: returns show little linear autocorrelation
 - ▶ but their **size** is highly predictable
-
- Stylised facts are the benchmark every model of returns must reproduce
 - ▶ the i.i.d. (independent and identically distributed) Normal-distribution model fails most of them

Data Used in This Chapter

Series	Price used	Type, currency	Calendar	Sample	T
S&P 500	daily close (plus open, high, low)	price index, USD	US trading days	1990–2026	9 245
BET-TR	daily close	total return, RON	BVB trading days	Sep 2014 – 2026	2 992
Bitcoin	daily close	price, USD	every day, 24/7	Sep 2014 – 2026	4 384
EUR/RON	BNR reference rate	RON per EUR, managed float	Romanian working days	Jul 2005 – 2026	5 352
Gold	daily close	spot, USD per ounce	weekdays only	1990–2026	9 559

- Daily closes up to 18 September 2026; BVB: Bucharest Stock Exchange
 - ▶ daily log returns on each series' own calendar
 - ▶ T : the number of daily returns in the series
- **Managed float:** a market-determined exchange rate with central-bank interventions
 - ▶ the official fixing of the National Bank of Romania (BNR) replaces the EUR/RON market (interbank) quotes
 - ▶ in several years these had a lag-1 autocorrelation of -0.4 to -0.7 , which came from quote errors
- In 1996–2005 the S&P 500 open equals the previous close on about 96% of days
 - ▶ still 16% in 2007
 - ▶ range-based estimators (built on the daily high–low range) therefore use data from 2008 on

Returns: Definitions and Aggregation (1/2)

	Simple return R_t	Log return r_t
Definition	$(P_t + D_t)/P_{t-1} - 1$	$\ln(1 + R_t) = \ln P_t - \ln P_{t-1}$ (no dividend)
Over k periods	$\prod_{j=0}^{k-1} (1 + R_{t-j}) - 1$	$\sum_{j=0}^{k-1} r_{t-j}$: additive over time
Across assets	$R_{p,t} = \sum_i w_i R_{i,t}$: exact	only approximately
Range	$R_t \geq -100\%$	$(-\infty, +\infty)$
Use	portfolios, performance reporting	single-asset dynamics, central limit theorem (CLT)

- ▣ The simple return is the relative change in the value of the investment; the log return is the change in the log price
- ▣ Notation
 - ▶ P_t : the closing price on day t ; D_t : the dividend paid in $(t-1, t]$
 - ▶ k : the number of periods aggregated; the index $j = 0, \dots, k-1$ runs over them
 - ▶ $R_{i,t}$: the simple return of asset i ; w_i : its weight in portfolio value, with $\sum_i w_i = 1$; $R_{p,t}$: the portfolio return
- ▣ With dividends use total-return or adjusted series (BET-TR)

Returns: Definitions and Aggregation (2/2)

- The log return is always below the simple return: $r = \ln(1 + R) < R$ for every $R \neq 0$
 - ▶ the reason: $\ln(1 + x) < x$ for every $x > -1$, $x \neq 0$, because the logarithm is concave
- The gap is negligible for small moves and large for big falls:
 - ▶ $R = +1\%$ gives $r = 0.995\%$
 - ▶ $R = -50\%$ gives $r = -69.3\%$
- Aggregation over time: the k -day log return is the sum of the k daily log returns
 - ▶ the k -day simple return needs the product of the growth factors $1 + R_{t-j}$
- Textbooks: [Campbell, Lo & MacKinlay \(1997\)](#), ch. 1; [Franke, Härdle & Hafner \(2019\)](#)

Annualisation, CAGR and Volatility Drag (1/2)

- ▣ Annualisation turns the daily mean and volatility into values per year
- ▣ Notation
 - ▶ A : the number of return intervals per year; 252 for stocks, 365 for crypto (traded every day), ≈ 260 for gold (the actual weekday quotes)
 - ▶ $\bar{r} = \frac{1}{T} \sum_{t=1}^T r_t$: the mean daily log return
 - ▶ $\hat{\sigma}$: the sample standard deviation of daily returns
- ▣ $\mu_{\text{ann}} = A \bar{r}$: the annual mean; needs only a constant mean
- ▣ $\sigma_{\text{ann}} = \sqrt{A} \hat{\sigma}$: the annual volatility; exact only for serially uncorrelated returns with constant variance
 - ▶ then daily variances add up, and the annual variance is $A \hat{\sigma}^2$
 - ▶ with autocorrelation the $\sqrt{A}$ factor is no longer correct
 - ▶ EUR/RON, lag-1 autocorrelation 0.17: annual volatility rises from 4.4% to 5.3% ([▶ Appendix](#); Seminar A7)

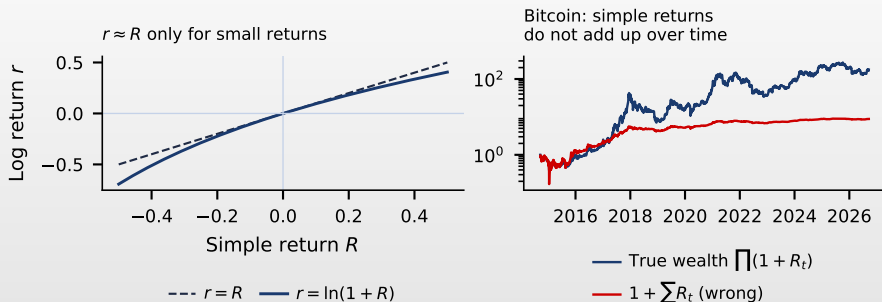
Annualisation, CAGR and Volatility Drag (2/2)

- **CAGR** (Compound Annual Growth Rate): the constant annual return that takes the price from P_0 to P_T

$$\text{CAGR} = (P_T/P_0)^{1/Y} - 1$$

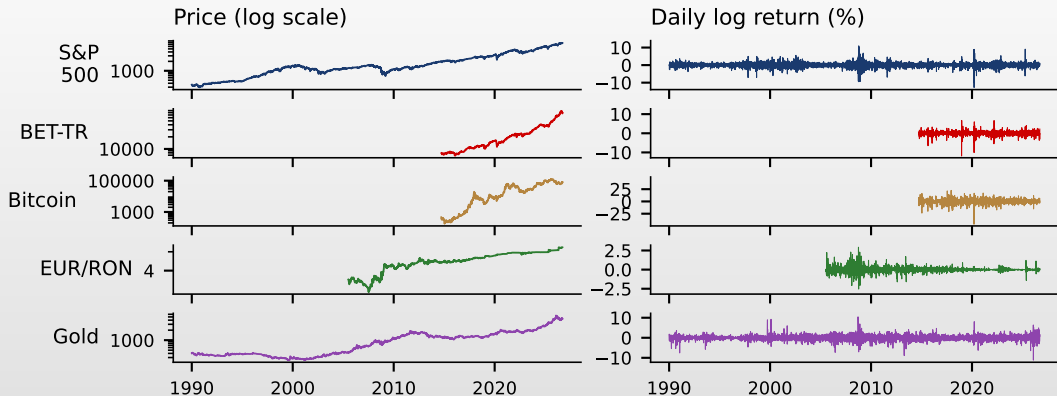
- ▶ P_0, P_T : the price at the start and at the end of the sample; Y : the number of years
- ▶ CAGR is the annual geometric mean of the growth factors
- The geometric mean $\approx \mu_{\text{arith}} - \sigma^2/2$
 - ▶ μ_{arith} : the annual arithmetic mean of simple returns; σ^2 : their annual variance
 - ▶ **volatility drag**: the gap between the arithmetic and the geometric mean, $\approx \sigma^2/2$; it grows with the square of volatility
- Examples
 - ▶ S&P 500 1990–2026: $\hat{\sigma} = 1.133\%$ a day, 18.0% a year; CAGR 8.69% (price index, no dividends)
 - ▶ Bitcoin 2014–2026: arithmetic mean 65.2% a year ($A = 365$), CAGR 53.9%
 - ▶ a drag of 11 pp (percentage points)

Simple and Log Returns in Practice



- Left: $r \approx R$ only for small moves
- Right: Bitcoin 2014–2026 multiplied wealth by 176.9, i.e. $\sum_t r_t = \ln 176.9 = 5.18$
 - ▶ the sum of daily log returns is $\ln(P_T/P_0)$, the log of the multiple
 - ▶ adding daily simple returns gives $1 + \sum_t R_t = 8.83$, 20 times below the true multiple

Five Markets: Prices and Returns (1/2)



Five Markets: Prices and Returns (2/2)

- **Left:** prices on log scale, trending and non-stationary
- **Right:** daily log returns fluctuate around a small mean
 - ▶ calm and turbulent periods alternate
 - ▶ extremes cluster (2008, March 2020)
- EUR/RON moves more than 1% on only 1.4% of days
- Bitcoin, 12 March 2020: log return -46.5%
 - ▶ the price is multiplied by e^{r_t} , so it falls by $1 - e^{-0.465} = 37.2\%$

Sample Moments and Their Sampling Error (1/3)

- ▣ Skewness and kurtosis describe the shape of the distribution, after removing its mean and scale

$$z_t = \frac{r_t - \bar{r}}{\hat{\sigma}}, \quad \hat{S} = \frac{1}{T} \sum_{t=1}^T z_t^3, \quad \hat{K} = \frac{1}{T} \sum_{t=1}^T z_t^4$$

- ▶ z_t : the standardised return, with mean 0 and standard deviation 1
 - ▶ $\hat{S}$: sample skewness; $\hat{S} < 0$: the left tail is longer, large losses dominate
 - ▶ $\hat{K}$: sample kurtosis; $\hat{K} > 3$: more mass in the centre and in the tails than under the Normal distribution
 - ▶ S and K without a hat: the population values; the hat marks an estimated value
- ▣ Normal distribution: $S = 0$, $K = 3$, excess kurtosis $K - 3 = 0$

Sample Moments and Their Sampling Error (2/3)

- **Standard error (SE):** the standard deviation of an estimator across samples
- Mean: $SE(\bar{r}) = \sigma/\sqrt{T}$, with σ the standard deviation of daily returns, estimated by $\hat{\sigma}$
 - ▶ the 95% confidence interval: $\bar{r} \pm 1.96 SE(\bar{r})$
 - ▶ daily means are 13–40 times smaller than $\hat{\sigma}$
 - ▶ so the mean is the least precise moment
 - ▶ S&P 500: $\bar{r} = 0.033\% \pm 0.023\%$ a day (95%), $T = 9,245$
- i.i.d. data from the Normal distribution: $\sqrt{T} \hat{S} \rightarrow N(0, 6)$ and $\sqrt{T}(\hat{K} - 3) \rightarrow N(0, 24)$
 - ▶ $\rightarrow$: convergence in distribution as $T \rightarrow \infty$; $N(0, v)$: the Normal distribution with mean 0 and variance v
 - ▶ so $SE(\hat{S}) \approx \sqrt{6/T}$ and $SE(\hat{K}) \approx \sqrt{24/T}$; for the S&P 500: 0.025 and 0.051

Sample Moments and Their Sampling Error (3/3)

- The **delta method** gives the standard error of a function of an estimator

$$\sqrt{T}(\hat{\theta} - \theta) \rightarrow N(0, \nu) \implies \sqrt{T}(g(\hat{\theta}) - g(\theta)) \rightarrow N(0, g'(\theta)^2 \nu)$$

- ▶ θ : a parameter, e.g. a moment; $\hat{\theta}$: its estimator; ν : the asymptotic variance of $\hat{\theta}$
- ▶ g : a differentiable function, e.g. the ratio of moments that defines $\hat{S}$; g' : its derivative
- ▶ the idea: the first-order Taylor expansion $g(\hat{\theta}) \approx g(\theta) + g'(\theta)(\hat{\theta} - \theta)$
- Any i.i.d. distribution: the delta method gives a Normal limit
 - ▶ for $\hat{S}$ only if $\mathbb{E}r^6 < \infty$, for $\hat{K}$ only if $\mathbb{E}r^8 < \infty$
 - ▶ $\mathbb{E}$: the expectation; $\mathbb{E}r^6 < \infty$: the sixth moment is finite
 - ▶ the variance of $\hat{S}$ involves the sixth moment, that of $\hat{K}$ the eighth
 - ▶ whether these moments exist is checked empirically in the section on tails

Three Classical Tests and Their Assumptions (1/3)

- ▣ **ARCH** (AutoRegressive Conditional Heteroskedasticity): today's variance depends on past returns
- ▣ **Volatility clustering**: days with large moves tend to follow days with large moves
- ▣ The **tail index** α measures how fast the probability of extreme values decays

$$P(|r_t| > x) \approx c x^{-\alpha} \quad \text{for large } x$$

- ▶ x : a positive threshold; $c > 0$: a scale constant
- ▶ a small α means a heavy tail: the probability decays slowly as x grows
- ▶ so $\mathbb{E}|r_t|^p < \infty$ only for $p < \alpha$, where p is the order of the moment
- ▶ the estimates in this chapter give $\alpha \approx 3$

Three Classical Tests and Their Assumptions (2/3)

- ▣ **JB** (Jarque–Bera): $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$
 - ▶ the distance of skewness and kurtosis from their Normal-distribution values
- ▣ **LB** (Ljung–Box): $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k)$
 - ▶ a weighted sum of the first m squared autocorrelations
 - ▶ $\hat{\rho}_k$: the sample autocorrelation at lag k , between r_t and r_{t-k} ; here $m = 10$ lags
- ▣ **ARCH-LM**: TR^2 from the regression of $\hat{\varepsilon}_t^2$ on $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-q}^2$
 - ▶ LM: the Lagrange multiplier test; $\hat{\varepsilon}_t = r_t - \bar{r}$: the residuals; $q = 5$ lags
 - ▶ R^2 : the share of the variation in $\hat{\varepsilon}_t^2$ explained by past squared residuals
- ▣ **Reading**: all three statistics are ≥ 0 , and large values reject the null H_0
 - ▶ reference distributions: $\chi_2^2, \chi_m^2, \chi_q^2, \chi_n^2$: the chi-squared distribution with n degrees of freedom
 - ▶ 5% critical values: $\chi_2^2 = 5.99, \chi_5^2 = 11.1, \chi_{10}^2 = 18.3$

Three Classical Tests and Their Assumptions (3/3)

Test	Statistic	H_0	χ^2 limit needs	Problem here
JB, Jarque & Bera (1980)	$T \left[\frac{\hat{\xi}^2}{6} + \frac{(\hat{\kappa}-3)^2}{24} \right]$	i.i.d., Normal distribution	H_0 itself (all moments exist)	a rejection does not say which part fails
LB, Ljung & Box (1978)	$T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$	$\rho_1 = \dots = \rho_m = 0$	i.i.d., not only uncorrelated	volatility clustering: over-rejects
ARCH-LM, Engle (1982)	$TR^2, \hat{\varepsilon}_t^2$ on q lags	i.i.d., no ARCH effects	$\mathbb{E}r^4 < \infty$	$\alpha \approx 3 < 4$: size unreliable

- The question is not how to run the tests
 - ▶ but whether their reference distribution holds for returns

Worked Example: JB for the S&P 500 (1/2)

- Test statistic: $JB = T \left(\frac{\hat{S}^2}{6} + \frac{(\hat{K} - 3)^2}{24} \right)$
 - ▶ it measures how far $\hat{S}$ and $\hat{K}$ are from their Normal-distribution values, 0 and 3
 - ▶ distributed χ^2_2 under the null (5% critical value: 5.99)
- S&P 500, 1990–2026: $T = 9,245$, $\hat{S} = -0.364$, $\hat{K} - 3 = 10.88$
 - ▶ skewness term $\hat{S}^2/6 = 0.132/6 = 0.022$
 - ▶ kurtosis term $10.88^2/24 = 118.4/24 = 4.93$
 - ▶ $JB = 9,245 \times (0.022 + 4.93) \approx 45,800$, against the critical value 5.99
- Almost all of the statistic comes from the kurtosis term
 - ▶ the problem is the tails, not the asymmetry

Worked Example: JB for the S&P 500 (2/2)

- Under the null all moments exist
 - ▶ so the p-value is valid for the **joint** null “i.i.d. and Normal distribution”
 - ▶ heavy tails make JB grow with T
 - ▶ this is power, not a broken p-value
- A rejection does not say which part fails
 - ▶ it may be the marginal distribution or independence
 - ▶ tests of the marginal distribution with HAC (heteroskedasticity and autocorrelation consistent) variances:
 - ▶ [Appendix](#)

Robust Alternative: Quantile-Based Skewness and Kurtosis (1/3)

- Two quantile-based measures [Kim & White \(2004\)](#), which need no moments
 - q_p : the quantile of order p of returns, the value below which a share p of days lies; $q_{0.5}$: the median
- Hinkley skewness** compares the lengths of the two tails around the median

$$\frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$$

- the numerator is $(q_{0.95} - q_{0.5}) - (q_{0.5} - q_{0.05})$: right tail minus left tail
- it lies in $[-1, 1]$; a negative value indicates a longer left tail
- Crow–Siddiqui excess kurtosis**, denoted K_{CS} , compares the width of the tails with the width of the centre

$$K_{CS} = \frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$$

- 2.91: the value of the ratio for the Normal distribution; a positive value indicates heavier tails
- both measures are 0 for the Normal distribution

Robust Alternative: Quantile-Based Skewness and Kurtosis (2/3)

- ▣ 95% confidence intervals (CI) from a **moving-block bootstrap**
 - ▶ it resamples blocks of 20 consecutive days (1,000 resamples)
 - ▶ the measure is recomputed on each resampled series
 - ▶ the blocks keep the dependence within them
- ▣ Valid for stationary, weakly dependent returns with a positive density at these quantiles
- ▣ A CI that excludes 0 indicates a significant departure from the Normal distribution

Robust Alternative: Quantile-Based Skewness and Kurtosis (3/3)

	Moment skew.	Hinkley [95% CI]	Excess kurt.	Crow–Siddiqui excess [95% CI]
S&P 500	−0.36	−0.064 [−0.092, −0.041]	10.9	1.54 [1.29, 1.79]
BET-TR	−1.40	−0.036 [−0.085, 0.025]	17.8	1.06 [0.73, 1.43]
Bitcoin	−0.70	−0.018 [−0.069, 0.032]	12.0	2.25 [1.88, 2.64]
EUR/RON	0.73	0.041 [−0.022, 0.093]	15.1	4.87 [3.74, 6.13]
Gold	−0.23	0.004 [−0.025, 0.030]	9.3	1.47 [1.29, 1.73]

- ▣ Quantile (Crow–Siddiqui) excess kurtosis is positive in all five markets
 - ▶ no CI contains 0
- ▣ Quantile-based negative skewness is established only for the S&P 500
 - ▶ BET-TR's moment skewness −1.40 comes from a few days (19 December 2018)

Summary Statistics: Five Markets

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Observations	9245	2992	4384	5352	9559
Mean (% per day)	0.033	0.077	0.118	0.007	0.025
Std. dev. (% per day)	1.13	0.98	3.49	0.28	1.01
Annualised volatility (%)	18.0	15.5	66.6	4.4	16.3
Skewness	-0.36	-1.40	-0.70	0.73	-0.23
Excess kurtosis	10.9	17.8	12.0	15.1	9.3
Min / max log return (%)	-12.8 / 11.0	-11.9 / 6.7	-46.5 / 22.5	-2.5 / 2.9	-11.1 / 10.4
Jarque-Bera	45805	40433	26482	51575	34732
Ljung-Box $Q(10)$, r_t	91.6	70.0	20.2	221.7	32.1
Ljung-Box $Q(10)$, r_t^2	8018	714	213	2420	1049
ARCH-LM (5 lags)	2229	371	115	871	483

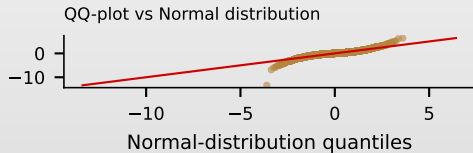
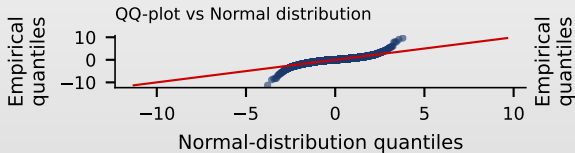
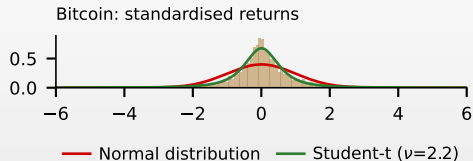
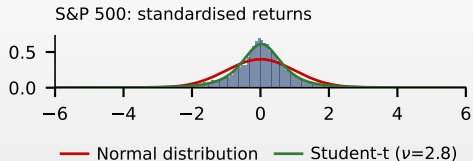
- Descriptive: heavy tails everywhere (excess kurtosis 9.3–17.8)
 - ▶ $Q(10)$ on r_t^2 is 10–90 times larger than on r_t
 - ▶ the size of returns is far more predictable than their direction
- Inferential: all i.i.d. p-values are below 0.03
 - ▶ but they are valid only under the assumptions in the table of the three classical tests

The Stylised Facts of Cont (2001)

Group	Facts
Distribution	2. heavy tails; 3. gain/loss asymmetry; 4. aggregational Gaussianity; 7. conditional heavy tails
Dependence of returns	1. absence of linear autocorrelation
Dependence of volatility	5. intermittency; 6. volatility clustering; 8. slow decay of the autocorrelation of $ r_t $; 11. asymmetry in time scales
Links to other variables	9. leverage effect; 10. volume/volatility correlation

- We re-test them on five markets, with data up to September 2026
 - we add the inference each test needs
- Early evidence: Mandelbrot (1963), Fama (1965); textbook treatment: Taylor (2008), Franke, Härdle & Hafner (2019)

Fact 2, Heavy Tails: Histograms and QQ-Plots (1/2)



Fact 2, Heavy Tails: Histograms and QQ-Plots (2/2)

- Extreme returns are far more frequent than under the Normal distribution
 - ▶ at the same variance
- Standardised returns $(r_t - \bar{r})/\hat{\sigma}$: higher peak and fatter tails
- QQ-plots compare empirical quantiles with those of the Normal distribution
 - ▶ they bend away from the line in both tails
- A Student- t fits much better

 MFM_ch1_heavy_tails

Tail Heaviness: Exceedances and the Student- t (1/3)

- The Student- t distribution, [Praetz \(1972\)](#): a heavy-tailed model for returns, with density

$$f(x) \propto \left(1 + \frac{(x - \mu)^2}{\nu s^2}\right)^{-(\nu+1)/2}$$

- Notation

- ▶ $f(x)$: the density at x ; $\propto$: “proportional to”, i.e. without the normalising constant
- ▶ μ : the centre of the distribution; $s > 0$: the scale parameter
- ▶ $\nu > 0$: the degrees of freedom; a small ν means heavy tails; as $\nu \rightarrow \infty$ we get the Normal distribution

- For a variable X with this distribution

- ▶ tail $P(X > x) \propto x^{-\nu}$: moments exist only of order $< \nu$
- ▶ variance $s^2 \nu / (\nu - 2)$ for $\nu > 2$
- ▶ excess kurtosis $6 / (\nu - 4)$ for $\nu > 4$

Tail Heaviness: Exceedances and the Student- t (2/3)

- $\hat{\nu}$, the estimated value of ν , is fitted by **maximum likelihood** to the whole distribution
 - ▶ it picks the μ , s and ν that make the observed data most probable
 - ▶ the many small returns dominate the fit
 - ▶ so a poor fit in the centre of the distribution moves $\hat{\nu}$
- EUR/RON: $\hat{\nu} = 1.2$ reflects that 57% of daily changes are below 0.1%, plus rare jumps

Tail Heaviness: Exceedances and the Student- t (3/3)

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Days beyond $3\hat{\sigma}$ (%)	1.55	1.44	1.82	2.06	1.42
Days beyond $5\hat{\sigma}$ (count)	30	12	10	29	26
Expected under the Normal distribution, $> 5\hat{\sigma}$	0.005	0.002	0.003	0.003	0.005
Student- t degrees of freedom $\hat{\nu}$ (maximum likelihood)	2.8	3.1	2.2	1.2	2.9

□ Under the Normal distribution

- ▶ 0.27% of days lie beyond 3σ (observed 5–8 times more)
- ▶ one 5σ day occurs every 6,900 years of trading (observed 10–30)

The Hill Estimator of the Tail Index (1/3)

- The **tail index** α , defined with the classical tests: $P(|X| > x) \approx c x^{-\alpha}$ for large x
 - ▶ X : the variable studied, here $|r_t|$; $\mathbb{E}|X|^p < \infty$ only for $p < \alpha$
 - ▶ $\alpha > 2$: finite variance; $\alpha > 4$: finite kurtosis
 - ▶ power laws were first argued by [Mandelbrot \(1963\)](#) and [Fama \(1965\)](#)
- The **Hill estimator**, [Hill \(1975\)](#): estimates α from the k largest observations only
 - ▶ order them from largest to smallest: $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)}$
 - ▶ $X_{(i)}$: the i -th largest value; n : the number of observations
 - ▶ k : the number of tail observations; $X_{(k+1)}$ acts as the threshold where the tail starts
- **Pareto tail**: $P(X > x) = (x/u)^{-\alpha}$ for $x \geq u$, with threshold u ; the exact power law

The Hill Estimator of the Tail Index (2/3)

- The Hill estimator is the inverse of the mean log exceedance over the threshold

$$\hat{\alpha}_k = \left[\frac{1}{k} \sum_{i=1}^k \ln X_{(i)} - \ln X_{(k+1)} \right]^{-1} = \left[\frac{1}{k} \sum_{i=1}^k \ln \frac{X_{(i)}}{X_{(k+1)}} \right]^{-1}$$

- ▶ $\ln(X_{(i)}/X_{(k+1)})$: how far the i -th value exceeds the threshold, on a log scale
- ▶ for a Pareto tail these exceedances average $1/\alpha$
- ▶ large exceedances mean a heavy tail, hence a small $\hat{\alpha}_k$
- Standard error $SE(\hat{\alpha}_k) \approx \hat{\alpha}_k/\sqrt{k}$; 95% CI: $\hat{\alpha}_k \pm 1.96 SE$
 - ▶ it assumes independent tail observations and a Pareto tail
 - ▶ clustering of extremes makes it larger (▶ Appendix)

The Hill Estimator of the Tail Index (3/3)

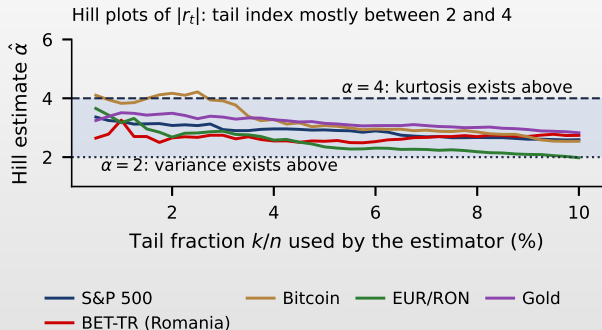
▣ Choice of k

- ▶ small k is noisy
- ▶ large k lets the centre in and biases $\hat{\alpha}$
- ▶ read the level where the **Hill plot** ($\hat{\alpha}_k$ against k) is flat, here $k = 1\text{--}5\%$ of n

▣ Comparison with the Student- t , whose tail index is $\alpha = \nu$

- ▶ EUR/RON: the Hill estimator gives $\hat{\alpha} = 2.8$ while $\hat{\nu} = 1.2$
- ▶ a Student- t fitted to the whole distribution misdescribes the tails

Hill Plots for Five Markets



- $\hat{\alpha}$ for $|r_t|$ at $k = 2.5\%$ of n
 - S&P 500 3.1, BET-TR 2.7, Bitcoin 4.2, EUR/RON 2.8, gold 3.3
- For $k = 1-5\%$: between 2.3 and 3.5 except Bitcoin (3.1–4.2)
- At a common threshold, loss tails are heavier than gain tails
 - BET-TR: 2.2 versus 3.1
 - Bitcoin: 2.7 versus 3.4
- Tails are modelled with extreme value theory ([Chapter 7](#))

Tail Indices with Confidence Intervals (1/2)

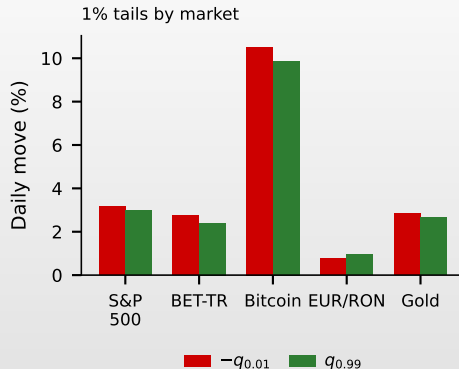
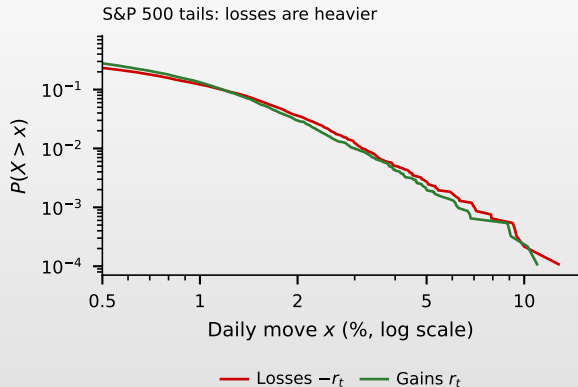
$ r_t $, $k = 2.5\%$ of n	n	k	$\hat{\alpha}$	SE i.i.d.	95% CI (i.i.d.)	SE block bootstrap
S&P 500	9245	231	3.07	0.20	[2.68, 3.47]	0.33
BET-TR	2992	74	2.67	0.31	[2.06, 3.28]	0.45
Bitcoin	4384	109	4.22	0.40	[3.43, 5.01]	0.47
EUR/RON	5352	133	2.82	0.24	[2.34, 3.30]	0.28
Gold	9559	238	3.31	0.21	[2.89, 3.73]	0.24

- Moving-block bootstrap (blocks of 20 days, 500 resamples) keeps the clustering of extremes
 - SEs are 1.1–1.6 times the i.i.d. ones

Tail Indices with Confidence Intervals (2/2)

- $\alpha = 4$ is outside every interval except Bitcoin's
 - ▶ also when the interval uses the bootstrap SE
 - ▶ the fourth moment very likely does not exist
- $\alpha = 2$ is outside every i.i.d. interval
 - ▶ with the bootstrap SE only BET-TR reaches it: the lower limit $\hat{\alpha} - 1.96 \text{ SE} = 2.67 - 1.96 \times 0.45 = 1.79$
- Bitcoin's 4.2 uses a threshold where its Hill plot is not yet flat
 - ▶ at a common threshold its two tails give 2.71 and 3.36 (▶ [Appendix](#))

Fact 3: Gain/Loss Asymmetry in the Data (1/4)

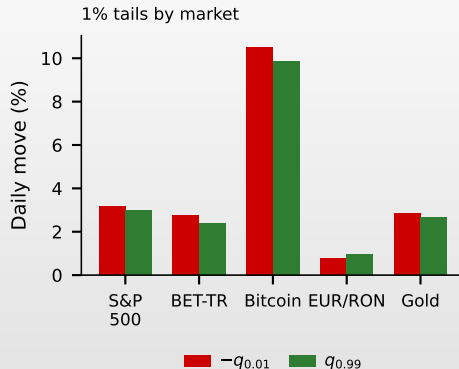
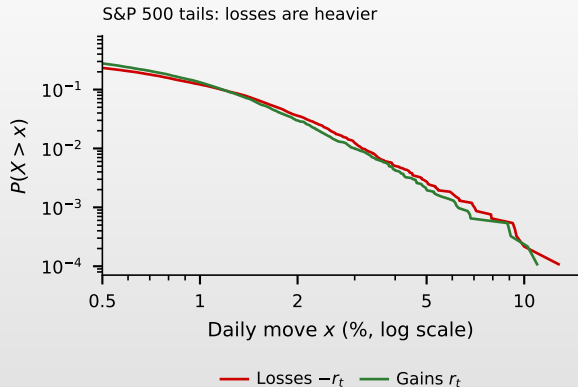


Fact 3: Gain/Loss Asymmetry in the Data (2/4)

- Survival functions $P(X > x)$ of losses and gains, log-log scale
 - ▶ $P(X > x)$: the share of days on which X exceeds x ; $X = -r_t$ for losses, $X = r_t$ for gains
 - ▶ large losses are more frequent than gains of the same size
- **VaR** (Value-at-Risk) 1% = $-q_{0.01}$, the daily loss threshold exceeded with probability 1%; $q_{0.01}$: the 1% quantile of returns (Chapter 7)

 MFM_ch1_heavy_tails

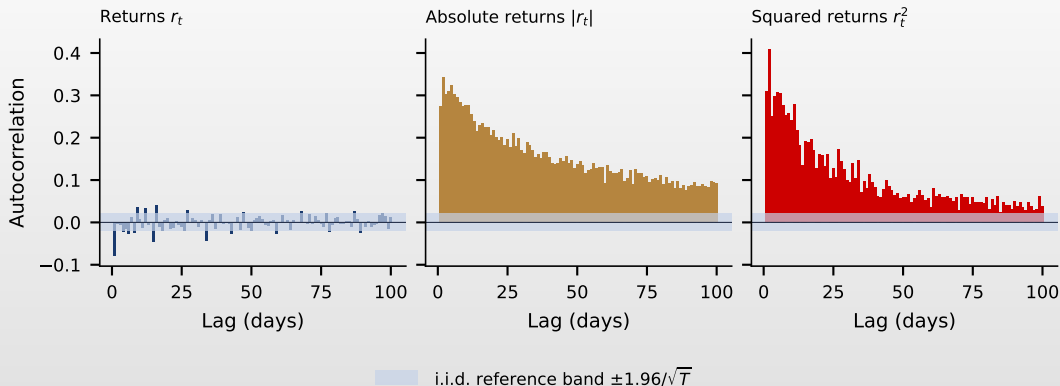
Fact 3: Gain/Loss Asymmetry in the Data (3/4)



Fact 3: Gain/Loss Asymmetry in the Data (4/4)

- VaR 1% exceeds the 99% return quantile $q_{0.99}$ for 4 of 5 markets
 - ▶ e.g. for the S&P 500: 3.15% versus 2.97%
- EUR/RON is the exception: leu depreciations (up-moves) are larger
 - ▶ $q_{0.99} = 0.94\%$ versus VaR 1% = 0.76%
- A Normal-distribution model with the right variance underestimates such quantiles
 - ▶ stress tests based on “ 3σ events” are too optimistic

Facts 1 and 6: Uncorrelated but Not Independent (1/4)

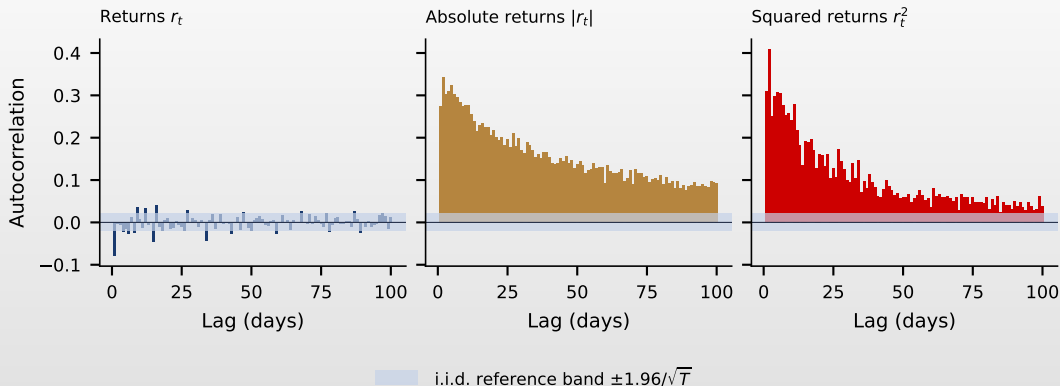


Facts 1 and 6: Uncorrelated but Not Independent (2/4)

- **Fact 1:** $\text{corr}(r_t, r_{t-k}) \approx 0$, where k is the lag in days
 - ▶ S&P 500 1990–2026, lag 1: -0.08
 - ▶ i.i.d. reference band: $\pm 1.96/\sqrt{T}$; outside it the autocorrelation is significant at 5%
- **Fact 6, volatility clustering:** $\text{corr}(|r_t|, |r_{t-k}|) > 0$ for many lags (lag 1: 0.27)
 - ▶ large moves follow large moves of either sign

 MFM_ch1_volatility_clustering

Facts 1 and 6: Uncorrelated but Not Independent (3/4)



Facts 1 and 6: Uncorrelated but Not Independent (4/4)

- Independence would require zero correlation of **every** function of returns
 - ▶ returns are uncorrelated but not independent
- Model $r_t = \sigma_t \varepsilon_t$ with σ_t driven by past returns
 - ▶ σ_t : the conditional volatility on day t ; ε_t : i.i.d. shocks with mean 0 and variance 1
 - ▶ GARCH, generalised ARCH ([Chapter 5](#))

 MFM_ch1_volatility_clustering

The ACF under Conditional Heteroskedasticity (1/3)

- ▣ **ACF** (AutoCorrelation Function): the sample autocorrelations $\hat{\rho}_k$, plotted against the lag k

$$\hat{\rho}_k = \frac{\sum_t x_t x_{t-k}}{\sum_t x_t^2}, \quad x_t = r_t - \bar{r}$$

- ▶ x_t : the demeaned return; $\hat{\rho}_k \in [-1, 1]$
- ▶ $\hat{\rho}_k > 0$: a day above the mean tends to be followed, k days later, by a day above the mean
- ▣ Uncorrelated returns whose conditional variance moves: $\hat{\rho}_k \approx N(0, \tau_k)$ [Romano & Thombs \(1996\)](#)
 - ▶ τ_k : the variance of $\hat{\rho}_k$, estimated by $\hat{\tau}_k = \sum_t x_t^2 x_{t-k}^2 / (\sum_t x_t^2)^2$
 - ▶ i.i.d.: $T\tau_k = 1$, which gives the classical band $\pm 1.96/\sqrt{T}$
 - ▶ volatility clustering: $T\tau_k > 1$; the classical band is too narrow and flags autocorrelations as significant too often

The ACF under Conditional Heteroskedasticity (2/3)

- ▣ Robust t -statistic: $\hat{\rho}_k / \sqrt{\hat{\tau}_k}$, compared with ± 1.96
- ▣ Robust portmanteau test [Lobato, Nankervis & Savin \(2001\)](#): null hypothesis $\rho_1 = \dots = \rho_m = 0$

$$\tilde{Q}(m) = \sum_{k=1}^m \frac{\hat{\rho}_k^2}{\hat{\tau}_k}$$

- ▶ the sum of the squared robust t -statistics for the first m lags, so $\tilde{Q}(m) \geq 0$
- ▶ reference distribution χ_m^2 ; large values reject the null
- ▶ its assumptions: [Appendix](#)
- ▣ The difference from the Ljung–Box $Q(m)$
 - ▶ $Q(m)$ divides each $\hat{\rho}_k^2$, approximately, by the i.i.d. variance $1/T$
 - ▶ $\tilde{Q}(m)$ divides by the estimated variance $\hat{\tau}_k$

The ACF under Conditional Heteroskedasticity (3/3)

Lag 1	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
$\hat{\rho}_1$	-0.079	0.061	-0.022	0.170	-0.011
$T\hat{\rho}_1$	5.00	4.35	2.89	6.85	2.43
t -statistic, i.i.d. / robust	-7.6 / -3.4	3.3 / 1.6	-1.5 / -0.9	12.4 / 4.7	-1.1 / -0.7

- S&P 500: $Q(10) = 91.6$ but $\tilde{Q}(10) = 19.0$ ($p = 0.041$)
- With robust SEs only the S&P 500 ($t = -3.4$) and EUR/RON ($t = 4.7$) keep a significant lag-1 autocorrelation
 - ▶ it is economically small

Heavy Tails from Mixtures of Normal Distributions (1/2)

- The model: the return is a Normal shock times a random volatility

$$r_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \sim N(0, 1) \text{ independent of } \sigma_t$$

- The kurtosis of returns is at least 3

$$K = \frac{\mathbb{E}[r_t^4]}{\mathbb{E}[r_t^2]^2} = 3 \frac{\mathbb{E}[\sigma_t^4]}{\mathbb{E}[\sigma_t^2]^2} \geq 3$$

- ▶ the middle equality: $\mathbb{E}[\varepsilon_t^4] = 3$ and $\mathbb{E}[\varepsilon_t^2] = 1$, and σ_t and ε_t are independent
- ▶ the inequality is Jensen's inequality ($\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$) with $X = \sigma_t^2$
- ▶ equality only if σ_t is constant; any variation in volatility creates excess kurtosis

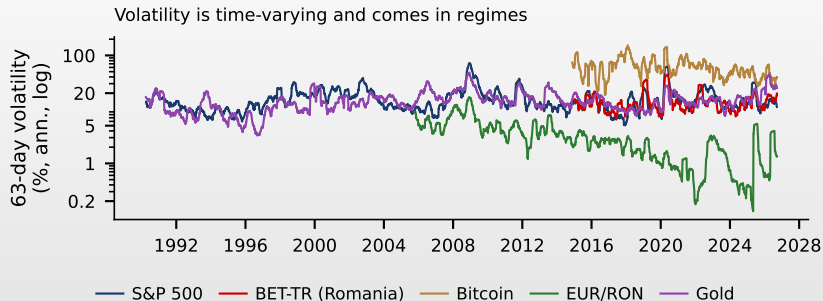
Heavy Tails from Mixtures of Normal Distributions (2/2)

- Example: $\sigma = 1\%$ on 90% of days and 3% on 10% of days

$$K = 3 \cdot \frac{0.9 \cdot 1 + 0.1 \cdot 81}{(0.9 + 0.1 \cdot 9)^2} = 3 \cdot \frac{9.0}{3.24} = 8.33$$

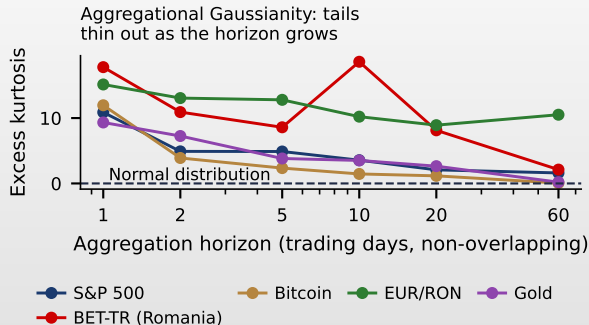
- ▶ numerator: $\mathbb{E}[\sigma^4] = 0.9 \cdot 1^4 + 0.1 \cdot 3^4$, in units of $(1\%)^4$
 - ▶ denominator: $\mathbb{E}[\sigma^2]^2$, with $\mathbb{E}[\sigma^2] = 0.9 \cdot 1^2 + 0.1 \cdot 3^2 = 1.8$
 - ▶ the excess kurtosis is 5.33, although each day's return, given σ , is Normally distributed
- This links heavy tails (Fact 2) to volatility clustering (Fact 6)
 - ▶ a mixture drawn independently every day has heavy tails but no clustering (Seminar A4, A9)

Fact 5: Intermittency in the Data



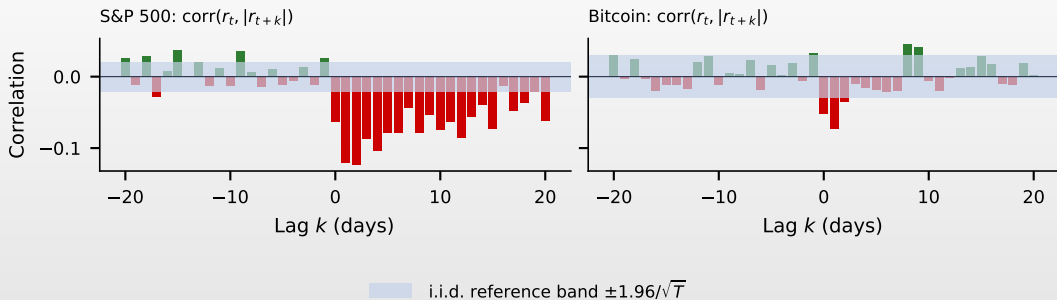
- Bursts of high volatility separated by calm periods: constant-volatility models cannot produce them
- 63-day rolling volatility $\sqrt{A} \times$ the standard deviation over the window, log scale
 - ▶ common spikes: 2008–2009, March 2020
- EUR/RON volatility collapsed after 2012, under the BNR's managed float

Fact 4: Aggregational Gaussianity



- h -day return: $r_t(h) = r_t + \dots + r_{t-h+1}$
 - ▶ computed in non-overlapping blocks
 - ▶ the CLT pulls sums towards the Normal distribution
 - ▶ slowly, because daily returns are dependent
- Excess kurtosis: S&P 500 $10.9 \rightarrow 1.6$ at $h = 60$, Bitcoin $12.0 \rightarrow 0.1$
- At $h = 60$ only 154 and 73 returns remain
 - ▶ SE of excess kurtosis (i.i.d., Normal): $\sqrt{24/n} = 0.39$ and 0.57
 - ▶ n : the number of h -day returns
 - ▶ so these values are very imprecise
- $h = 60$ is a quarter for the S&P 500 but two months for Bitcoin

Fact 9: The Leverage Effect (1/3)

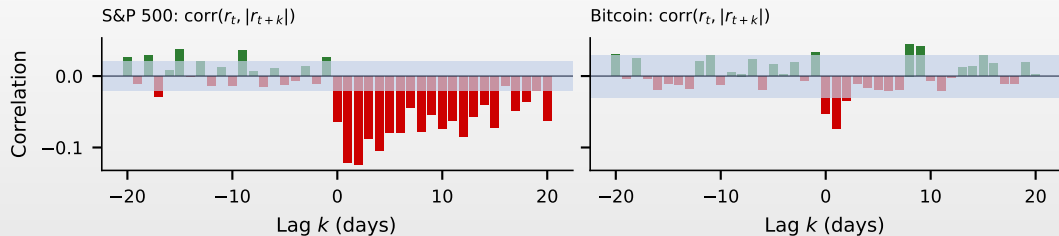


Fact 9: The Leverage Effect (2/3)

- $L(k) = \text{corr}(r_t, |r_{t+k}|)$, $k > 0$: the correlation of today's return with the size of the return k days later
 - ▶ $L(k) < 0$: after falls, returns become larger in absolute value, i.e. volatility rises
- S&P 500: $L(1) = -0.12$, still -0.07 at $k = 10$
 - ▶ no effect for $k < 0$
- BET-TR -0.12 ; weak for Bitcoin (-0.07)
 - ▶ negligible for gold (-0.02), reversed for EUR/RON ($+0.10$)

 MFM_ch1_leverage_volume

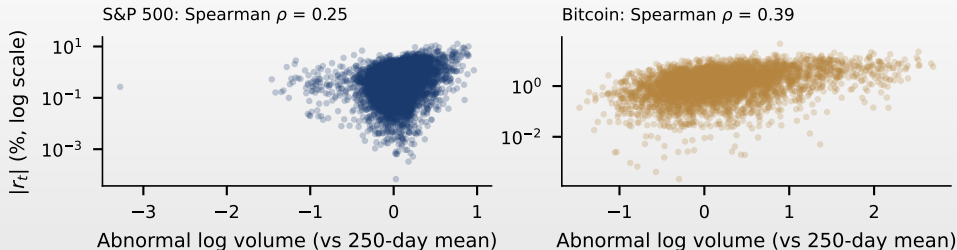
Fact 9: The Leverage Effect (3/3)



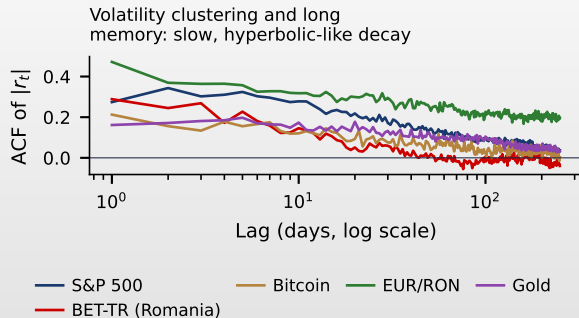
i.i.d. reference band $\pm 1.96/\sqrt{T}$

- Possible explanations
 - ▶ financial leverage of firms [Christie \(1982\)](#)
 - ▶ the retarded volatility model [Bouchaud, Matacz & Potters \(2001\)](#)
- Asymmetric GARCH models: GJR-GARCH, EGARCH ([Chapter 5](#))

Fact 10: Volume and Volatility Move Together

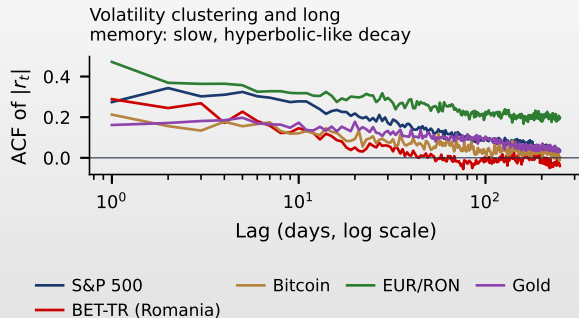


- Abnormal log volume ($\ln V_t$ minus its 250-day moving average) against $|r_t|$; V_t : the volume traded on day t
 - Spearman rank correlation, the correlation between the ranks of the two series, in $[-1, 1]$: 0.25 (S&P 500) and 0.39 (Bitcoin)
- Mixture-of-distributions hypothesis [Clark \(1973\)](#): information flow drives both price moves and volume
 - survey of the empirical evidence: [Karpoff \(1987\)](#)
- Bitcoin volume is exchange-reported
 - it is partly **wash trading** (an exchange trading with itself) [Cong, Li, Tang & Yang \(2023\)](#)

Fact 8: Slow Decay of the ACF of $|r_t|$ (1/2)

- ρ_k : the autocorrelation of $|r_t|$ at lag k ; $c > 0$: a constant
- **Short memory:** $\sum_k |\rho_k| < \infty$
 - ▶ e.g. exponential decay $\rho_k \approx c \phi^k$, with $0 < \phi < 1$ (GARCH(1,1), Chapter 5)
- **Long memory:** hyperbolic decay $\rho_k \approx c k^{2d-1}$, $0 < d < \frac{1}{2}$ Ding, Granger & Engle (1993)
 - ▶ d : the memory parameter; a larger d means slower decay, and $\sum_k \rho_k = \infty$

Fact 8: Slow Decay of the ACF of $|r_t|$ (2/2)



- S&P 500: 0.27 at lag 1, 0.22 at lag 20, still 0.09 at lag 100
 - ▶ BET-TR decays fastest
- Slow empirical decay is not yet established long memory
 - ▶ structural breaks, i.e. abrupt shifts in the level of volatility, can imitate it

Estimating the Memory Parameter d (1/2)

- Long memory in the frequency domain: the spectral density grows without bound at low frequencies

$$f(\lambda) \approx G\lambda^{-2d} \quad \text{as } \lambda \rightarrow 0$$

- ▶ $f(\lambda)$: the spectral density, the contribution of cycles of frequency λ to the variance of the series; small λ : long cycles
- ▶ $G > 0$: a constant; d : the memory parameter
- ▶ semiparametric estimators use only the m lowest Fourier frequencies $\lambda_j = 2\pi j/T$, $j = 1, \dots, m$

- Two estimators (formulas in the [Appendix](#))

- ▶ local Whittle [Robinson \(1995\)](#): $\sqrt{m}(\hat{d} - d) \rightarrow N(0, \frac{1}{4})$, so $\text{SE} = 1/(2\sqrt{m})$
- ▶ log-periodogram regression [Geweke & Porter-Hudak \(1983\)](#): $\text{SE} = \pi/\sqrt{24m}$
- ▶ here $m = T^{0.65}$

Estimating the Memory Parameter d (2/2)

$ r_t , m = T^{0.65}$	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
m	378	181	232	265	386
Local Whittle $\hat{d}$ (SE)	0.48 (0.03)	0.37 (0.04)	0.35 (0.03)	0.39 (0.03)	0.40 (0.03)
Log-periodogram $\hat{d}$ (SE)	0.50 (0.03)	0.41 (0.05)	0.35 (0.04)	0.38 (0.04)	0.42 (0.03)

- $\hat{d} \approx 0.35\text{--}0.50$ everywhere
 - ▶ at least 8 SEs from $d = 0$ (short memory) with both estimators
- The S&P 500 is close to the stationarity boundary $\frac{1}{2}$
 - ▶ for $d \geq \frac{1}{2}$ the series is no longer stationary

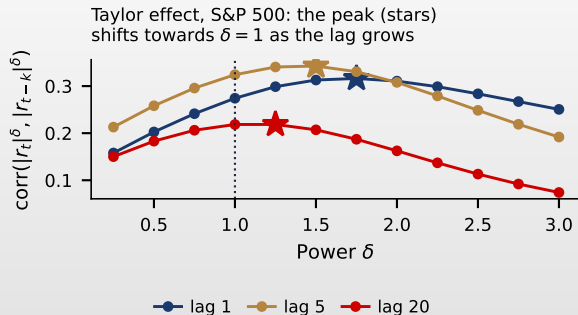
Long Memory or Structural Breaks?

- Rare shifts in the level of volatility produce a slowly decaying ACF of $|r_t|$
 - ▶ and $\hat{d} > 0$ without long memory [Diebold & Inoue \(2001\)](#), [Mikosch & Stărică \(2004\)](#)
- Check: locate one variance break with the cumulative sum of squares [Inclán & Tiao \(1994\)](#)
 - ▶ then re-estimate d (local Whittle) on each side

$ r_t $	Break date	Volatility before / after (%)	$\hat{d}$ before (SE)	$\hat{d}$ after (SE)
S&P 500	20 Jul 1998	12.6 / 19.3	0.28 (0.04)	0.51 (0.03)
BET-TR	17 Dec 2018	12.5 / 16.9	0.19 (0.05)	0.37 (0.04)
Bitcoin	19 Jun 2022	74.9 / 47.7	0.39 (0.04)	0.15 (0.05)
EUR/RON	27 Jun 2013	6.6 / 2.2	0.35 (0.04)	0.35 (0.04)
Gold	29 Nov 2005	13.2 / 18.3	0.30 (0.03)	0.37 (0.03)

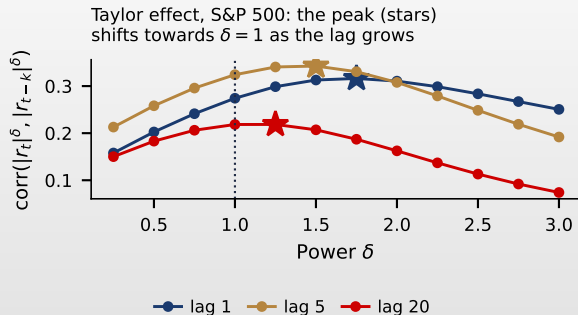
- $\hat{d}$ stays well above 0 in every sub-sample
 - ▶ one break does not explain the memory
- But $\hat{d}$ moves (Bitcoin $0.39 \rightarrow 0.15$)
 - ▶ several breaks or a time-varying d remain open
 - ▶ the break dates are descriptive

The Taylor Effect (1/2)



- **Taylor effect:** absolute returns $|r_t|$ are more strongly autocorrelated than squared returns r_t^2
 - ▶ $\text{corr}(|r_t|^\delta, |r_{t-k}|^\delta)$ peaks at a power δ close to 1
 - ▶ not at $\delta = 2$
- The chart: this autocorrelation as a function of $\delta > 0$
 - ▶ $\delta = 1$: absolute returns
 - ▶ $\delta = 2$: squared returns

The Taylor Effect (2/2)



- S&P 500: maximum at $\delta = 1.75$ (lag 1), 1.5 (lag 5) and 1.25 (lag 20)
 - ▶ the optimum moves towards 1 at longer lags
 - ▶ and is never at $\delta = 2$
- Documented by Taylor (1986), named by Granger & Ding (1995)
- The effect motivates
 - ▶ power-GARCH models, which model σ_t^δ with an estimated δ (Chapter 5)
 - ▶ absolute-value volatility measures

Moments, Tail Index and Valid Inference

Statistic	Needs	With Hill $\hat{\alpha} \approx 2.5-3.5$
Mean, variance of r_t	$\alpha > 2$	finite, slow convergence
ACF of $ r_t $, robust ACF bands, $\tilde{Q}$	$\alpha > 4$ for Normal limits	doubtful
Kurtosis, ACF of r_t^2 , ARCH-LM, Q on r_t^2	$\alpha > 4$ to exist, $\alpha > 8$ for Normal limits	no standard inference
Skewness	$\alpha > 6$ for a Normal limit	no standard inference
Quantiles, number of tail days, Hill $\hat{\alpha}$	no moments	valid, with dependence-robust SEs

- With an infinite fourth moment the sample ACF of r_t^2 converges slowly
 - ▶ its limit is not the Normal distribution [Davis & Mikosch \(1998\)](#)
 - ▶ the $\pm 1.96/\sqrt{T}$ band and the p-values of Q on r_t^2 and ARCH-LM are not reliable
- The conclusion “volatility clusters” is robust
 - ▶ the exact magnitudes are not robust
 - ▶ one reason behind the Taylor effect: $|r_t|$ needs fewer finite moments than r_t^2

Facts 7 and 11: Conditional Tails and Time Scales

Fact 7: conditional heavy tails

- Fit a volatility model, e.g. GARCH ([Chapter 5](#))
 - ▶ and standardise: $\hat{z}_t = r_t / \hat{\sigma}_t$, with $\hat{\sigma}_t$ the model's volatility estimate for day t
- The kurtosis of $\hat{z}_t$ is lower than that of r_t , still above 3
- Models need heavy-tailed innovations (Student- t , possibly skewed)

Fact 11: asymmetry in time scales

- Weekly volatility predicts daily volatility better than the reverse
- Traders with different horizons react at different frequencies
- The heterogeneous autoregressive model of realised volatility (**HAR-RV**, [Chapter 9](#))
 - ▶ realised volatility is the sum of squared intraday returns

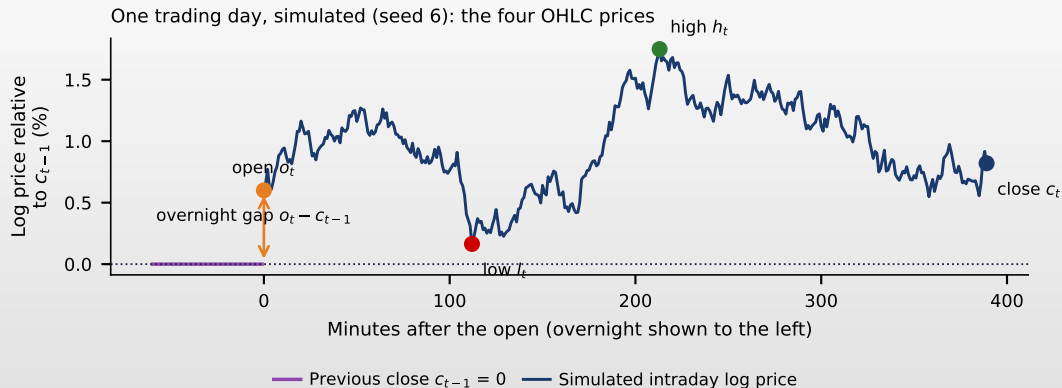
- Both facts are documented in [Cont \(2001\)](#)
- Returns standardised by realised volatility from intraday data are close to the Normal distribution
 - ▶ [Andersen, Bollerslev, Diebold & Ebens \(2001\)](#) ([Chapter 9](#))

Differences between the Five Markets

Market	Institutional feature	What differs	Candidate explanation
Bitcoin	trades 24/7; $A = 365$	volatility 66.6% versus 18.0% (S&P 500); worst day -46.5% ; weak leverage (-0.07); ACF of $ r_t $ at lag 100: 0.02 versus 0.09	no corporate leverage; volatility rises after large moves of either sign
EUR/RON	BNR daily fixing, managed float	volatility 4.4% but excess kurtosis 15.1; $\hat{\rho}_1 = +0.17$; largest moves October 2008 ($+2.9\%$, -2.5%)	adjustments smoothed over several days
BET-TR	total return, frontier/emerging market	skewness -1.40 ; worst day -11.9% on 19 Dec 2018 (bank-asset tax announced on 18 Dec, Government Emergency Ordinance 114/2018); $\hat{\rho}_1 = +0.06$	policy shocks; non-synchronous trading of constituents Campbell, Lo & MacKinlay (1997), ch. 3

- The explanations are hypotheses, not identified causes
- Low volatility is not low tail risk
- Heavy tails and volatility clustering hold in all five markets
 - ▶ autocorrelation and asymmetry depend on institutions

Open, High, Low and Close (OHLC) Prices (1/3)



Open, High, Low and Close (OHLC) Prices (2/3)

- **OHLC:** four prices of day t
 - ▶ O_t : the open
 - ▶ H_t : the high
 - ▶ L_t : the low
 - ▶ C_t : the close
- **Range-based** estimators use the whole path of the day
 - ▶ the range H_t/L_t is large on volatile days
 - ▶ even if the close equals the open

Open, High, Low and Close (OHLC) Prices (3/3)

- Logs of the four prices

$$o_t = \ln O_t, \quad h_t = \ln H_t, \quad l_t = \ln L_t, \quad c_t = \ln C_t$$

- ▶ the lower-case letter is the log of the price with the same initial

- Log range of day t

$$h_t - l_t = \ln(H_t/L_t) \geq 0$$

- ▶ approximately the percentage gap between the day's high and low

- Three log returns of day t

- ▶ close-to-close: $c_t - c_{t-1}$
- ▶ overnight: $o_t - c_{t-1}$
- ▶ intraday: $c_t - o_t$

- **Close-to-close estimator:** the standard deviation of $c_t - c_{t-1}$

- ▶ one observation per day

- **Efficiency** of an estimator X : $\text{Var}(\hat{\sigma}_{CC}^2)/\text{Var}(\hat{\sigma}_X^2)$

- ▶ an efficiency of 5 means the same precision with 5 times fewer days

Parkinson and Garman–Klass Estimators (1/2)

- ▣ **Parkinson, Parkinson (1980)**: estimates the daily variance from the daily range only

$$\hat{\sigma}_P^2 = \frac{1}{4 \ln 2} \frac{1}{n} \sum_{t=1}^n (h_t - l_t)^2$$

- ▶ n : the number of days; $h_t - l_t$: the log range of day t
 - ▶ for a Brownian motion with daily variance σ^2 , $\mathbb{E}[(h_t - l_t)^2] = 4 \ln 2 \sigma^2$; dividing by $4 \ln 2$ makes the estimator unbiased
 - ▶ $\hat{\sigma}_P^2$ is a daily variance; the annual volatility is $\sqrt{A \hat{\sigma}_P^2}$
- ▣ Assumption: a Brownian motion without drift and without overnight jumps
 - ▶ Brownian motion: the continuous-time random walk ([Chapter 11](#))
 - ▶ efficiency relative to close-to-close: about 5 (Seminar A8); the same precision from 5 times fewer days

Parkinson and Garman–Klass Estimators (2/2)

- ▣ **Garman–Klass**, Garman & Klass (1980): combines the range with the open-to-close move

$$\hat{\sigma}_{GK}^2 = \frac{1}{n} \sum_{t=1}^n \left[\frac{1}{2} (h_t - l_t)^2 - (2 \ln 2 - 1) (c_t - o_t)^2 \right]$$

- ▶ $c_t - o_t$: the open-to-close log return
 - ▶ the weights $\frac{1}{2}$ and $2 \ln 2 - 1 \approx 0.386$ come from the minimum-variance estimator derived by Garman and Klass
 - ▶ the same assumptions as Parkinson: no drift, no overnight jumps
- ▣ Efficiency relative to close-to-close: 7–8

Rogers–Satchell and Yang–Zhang Estimators (1/2)

- ▣ **Rogers–Satchell**, [Rogers & Satchell \(1991\)](#): unbiased under a non-zero **drift** (expected trend of the log price)

$$\hat{\sigma}_{RS}^2 = \frac{1}{n} \sum_{t=1}^n [(h_t - o_t)(h_t - c_t) + (l_t - o_t)(l_t - c_t)]$$

- ▶ the terms measure how far the high and the low moved away from the open and from the close
- ▶ both products are ≥ 0 , because $l_t \leq o_t, c_t \leq h_t$
- ▶ for a Brownian motion with drift their mean is exactly the daily variance
- ▶ it does not include the overnight jump

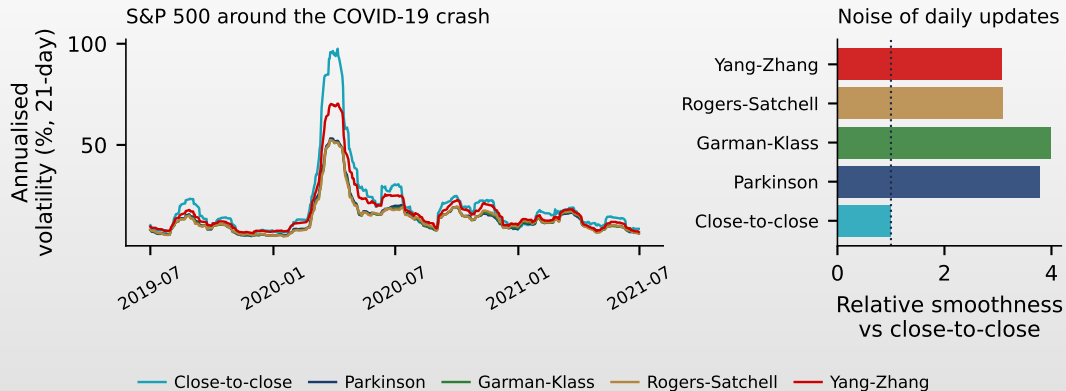
Rogers–Satchell and Yang–Zhang Estimators (2/2)

- ▣ **Yang–Zhang, Yang & Zhang (2000)**: adds the overnight (close-to-open) jump

$$\hat{\sigma}_{YZ}^2 = \hat{\sigma}_{\text{overnight}}^2 + k \hat{\sigma}_{\text{open-close}}^2 + (1 - k) \hat{\sigma}_{RS}^2, \quad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

- ▶ $\hat{\sigma}_{\text{overnight}}^2$: the sample variance of $o_t - c_{t-1}$, the overnight return
 - ▶ $\hat{\sigma}_{\text{open-close}}^2$: the sample variance of $c_t - o_t$
 - ▶ k : the weight, chosen to make the combination minimum-variance; $k \approx 0.145$ for large n
 - ▶ minimum-variance combination among drift-independent estimators robust to opening jumps
- ▣ Range estimators are the bridge to realised volatility from intraday data ([Chapter 9](#))

Estimators on the S&P 500 (1/2)

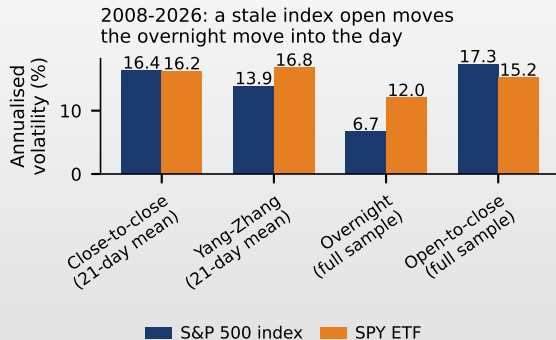


Estimators on the S&P 500 (2/2)

- 21-day windows, 2008–2026; averages
 - ▶ close-to-close 16.4%, Yang–Zhang 13.9%, Parkinson 13.0%, Garman–Klass 12.4%, Rogers–Satchell 12.3%
- Pure range estimators are lower: they miss overnight gaps
- The range-based series are 3–4 times smoother: a property of the estimates, not by itself proof of efficiency

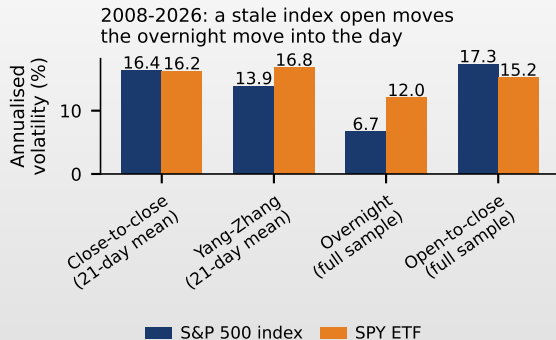
 MFM_ch1_vol_estimators

Stale Opens: the S&P 500 Index versus SPY (1/2)



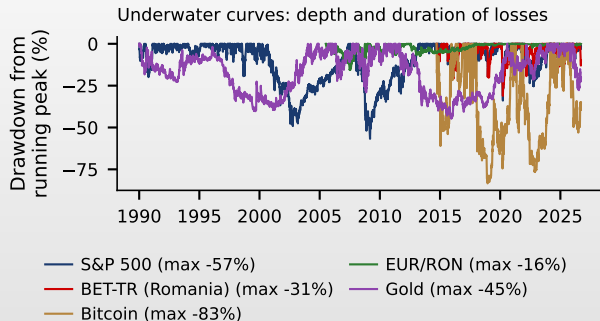
- **Stale open:** the index open is computed from the constituents' first prices
 - ▶ at 9:30 many stocks have not traded yet
 - ▶ so their previous close enters the open
- SPY, an ETF (exchange-traded fund) that tracks the S&P 500, opens at a traded price
- Open = previous close
 - ▶ about 96% of days in 1996–2005
 - ▶ 3.5% over 2008–2026 (up to 26% in 2012–2013)

Stale Opens: the S&P 500 Index versus SPY (2/2)



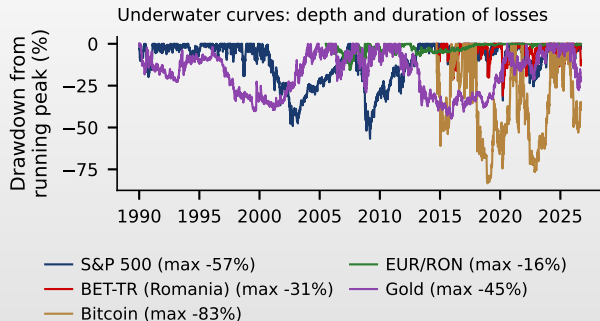
- A stale open moves part of the overnight move into the day
 - ▶ index overnight volatility: 6.7%
 - ▶ SPY, whose open is a traded price: 12.0%
- Yang-Zhang falls below close-to-close on the index only
 - ▶ use a traded instrument for estimators that use the open

Drawdowns Across Markets (1/2)



- Drawdown $DD_t = P_t / \max_{s \leq t} P_s - 1$: the loss from the peak reached up to day t ([Chapter 0](#))
 - ▶ $\max_{s \leq t} P_s$: the highest price up to day t ;
 $DD_t \in [-1, 0]$, with 0 at a new peak
 - ▶ maximum drawdown $MDD = \min_t DD_t$: the deepest fall from a peak
- MDD in the five markets
 - ▶ Bitcoin -83%, S&P 500 -57%, gold -45%
 - ▶ BET-TR -31% (March 2020), EUR/RON -16%

Drawdowns Across Markets (2/2)



- S&P 500: 9 October 2007 (1,565.15) to 9 March 2009 (676.53): -56.8%

- ▶ **recovery**: the price regaining its previous peak
- ▶ after a fall of x it needs a gain of $1/(1-x) - 1$
- ▶ here $1/(1-0.568) - 1 = +131\%$

- The MDD grows with the sample length

- ▶ compare only equal horizons (▶ [Appendix](#))

Recovery Duration after the Maximum Drawdown

- ▣ **Recovery:** the price regaining the peak it had before the fall
- ▣ **Recovery duration:** the time from that peak until the price reaches it again

	MDD	Peak	Trough	Peak regained
S&P 500	−57%	Oct 2007	Mar 2009	Mar 2013 (5.5 years)
BET-TR	−31%	Jan 2020	Mar 2020	Dec 2020 (0.9 years)
Bitcoin	−83%	Dec 2017	Dec 2018	Nov 2020 (3.0 years)
EUR/RON	−16%	Dec 2005	Jul 2007	Jan 2008 (2.0 years)
Gold	−45%	Sep 2011	Dec 2015	Jul 2020 (8.9 years)

- ▣ Depth and duration are different risks
 - ▶ gold's drawdown was shallower than Bitcoin's but lasted three times longer
- ▣ Fast V-shaped recoveries (BET-TR 2020)
 - ▶ versus long underwater periods (gold 2011–2020)
- ▣ For EUR/RON a “drawdown” is a fall of the euro against the leu

Sharpe, Sortino and Calmar Ratios (1/3)

▣ **Sharpe ratio** Sharpe (1966), Sharpe (1994)

- ▶ the excess return earned per unit of volatility

$$SR = \frac{\mu_{\text{ann}} - r_f}{\sigma_{\text{ann}}}$$

- ▶ r_f : the risk-free rate; $r_f = 0$ here, for comparability
 - ▶ $\mu_{\text{ann}} = A\bar{R}$: the annual mean of **simple** returns
 - ▶ $\bar{R}$: the mean daily simple return
 - ▶ σ_{ann} : the annual volatility of simple returns
 - ▶ a higher SR means more return for the same risk
- ▣ S&P 500 1990–2026: $\mu_{\text{ann}} = 9.95\%$, $\sigma_{\text{ann}} = 17.97\%$, $SR = 0.55$
- ▶ $\mu_{\text{ann}} = \text{mean log return } 8.33\% \text{ plus } \approx \sigma^2/2 = 1.62 \text{ pp}$

Sharpe, Sortino and Calmar Ratios (2/3)

▣ **Sortino ratio** Sortino & van der Meer (1991)

- ▶ the same numerator as the Sharpe ratio
- ▶ in the denominator: only the volatility of losses

$$SoR = \frac{\mu_{\text{ann}} - r_f}{\sigma_{\text{ann}}^-}, \quad \sigma^- = \sqrt{\frac{1}{T} \sum_{t=1}^T \min(R_t, 0)^2}$$

- ▶ σ^- : the daily downside deviation
- ▶ $\min(R_t, 0)$ keeps only negative returns
- ▶ the average runs over all T days; positive days count as 0
- ▶ $\sigma_{\text{ann}}^- = \sqrt{A} \sigma^-$: the annual version

- ▣ S&P 500: $\sigma_{\text{ann}}^- = 12.68\%$, $SoR = 0.78$

Sharpe, Sortino and Calmar Ratios (3/3)

▣ **Calmar ratio**

- ▶ the annual growth earned per unit of maximum drawdown

$$\text{Calmar} = \frac{\text{CAGR}}{|MDD|}$$

- ▶ CAGR: the compound annual growth rate
- ▶ MDD: the maximum drawdown in the sample

▣ S&P 500: $8.69\%/56.78\% = 0.15$

- ▶ the value is driven by one episode

Sharpe Ratios with Standard Errors (1/2)

□ Delta method, i.i.d. returns from the Normal distribution [Lo \(2002\)](#)

- ▶ SR_d : the daily Sharpe ratio; $\widehat{SR}_d$: its estimate; $\text{Var}(\widehat{SR}_d) \approx (1 + SR_d^2/2)/T$
- ▶ annualised, $SR_{\text{ann}} = \sqrt{A} SR_d$: $\text{SE}(\widehat{SR}_{\text{ann}}) \approx \sqrt{A(1 + SR_d^2/2)/T} \approx 1/\sqrt{Y}$, with $Y = T/A$ years
- ▶ the span matters, not the frequency

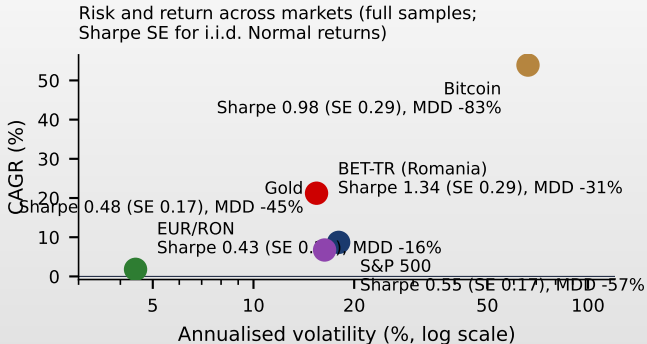
	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Years Y	36.7	12.0	12.0	21.2	36.7
Sharpe ratio (SE, Normal)	0.55 (0.17)	1.34 (0.29)	0.98 (0.29)	0.43 (0.22)	0.48 (0.17)

 MFM_ch1_risk_measures

Sharpe Ratios with Standard Errors (2/2)

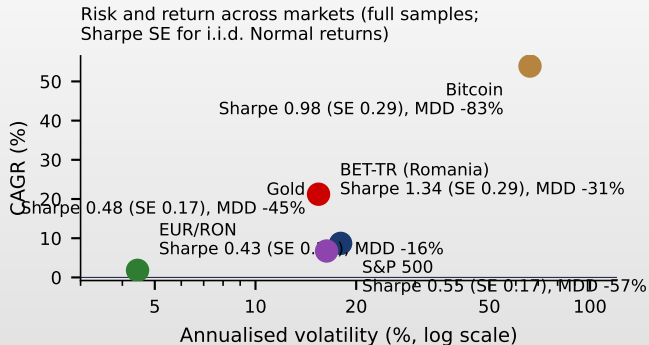
- Clustering: HAC or studentised block-bootstrap tests [Ledoit & Wolf \(2008\)](#)
 - ▶ they still need finite fourth moments, which the Hill estimates put in doubt
- BET-TR versus S&P 500, common days 2014–2026
 - ▶ difference 0.61, $z = 1.75$ ($p = 0.08$)
 - ▶ z : the difference divided by its standard error, compared with $N(0, 1)$
 - ▶ block-bootstrap 95% CI $[-0.01, 1.21]$ (Seminar B14)
- Calmar has no usable sampling theory
 - ▶ its drawdown grows with the sample length

Risk and Return Across Markets (1/2)



- Labels: the Sharpe ratio and, in brackets, its standard error (SE)
 - ▶ $SE = \sqrt{A(1 + SR_d^2)/T}$
 - ▶ valid for i.i.d. Normal simple returns
 - ▶ full table in the [Appendix](#)
- Different sample periods: not a ranking

Risk and Return Across Markets (2/2)



- S&P 500 without dividends; BET-TR with dividends and in RON
 - ▶ RON short-term rates were well above USD rates
 - ▶ with $r_f = 0$ the BET-TR Sharpe ratio is inflated
- Heavy tails and negative skewness flatter the Sharpe ratio
 - ▶ report it with drawdowns
 - ▶ and with tail-risk measures: VaR 1% and the tail index $\hat{\alpha}$

Case Study (1/14): Research Question and Dependent Variables

- Paper: Guyon & Lekeufack (2023), *Quantitative Finance*
 - ▶ the paper builds on Bollerslev, Hood, Huss & Pedersen (2018)
- Research question
 - ▶ what share of an equity index's volatility is explained by its past daily returns alone?
 - ▶ what functional form does this dependence take?
- Three dependent variables, modelled separately (Sections 3.4–3.5 of the paper)
 - ▶ VIX: the implied volatility of the S&P 500 over the next 30 days, computed from option prices
 - ▶ VIX9D: the same measure over a 9-day horizon
 - ▶ next-day realised volatility, computed from 5-minute returns

Case Study (2/14): Requirement, Relevance and Data

- Requirement: a single model must reproduce simultaneously
 - ▶ volatility clustering (Fact 6)
 - ▶ the slow decay of the autocorrelation of $|r_t|$ (Fact 8)
 - ▶ the leverage effect (Fact 9)
- Why it matters: if volatility is mostly a function of the price path, it is mostly endogenous
 - ▶ a **path-dependent volatility (PDV)** model needs little extra randomness
 - ▶ unlike stochastic volatility models ([Chapters 11–12](#))
- Data: daily closes of the S&P 500, Euro Stoxx 50, FTSE 100, DAX and Nikkei 225
 - ▶ training 1 Jan 2000 – 31 Dec 2018
 - ▶ test 1 Jan 2019 – 15 May 2022

Case Study (3/14): Model

- Equation (3.7) in the paper: today's volatility is a linear function of two summaries of past returns

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \Sigma_t$$

- Notation

- ▶ σ_t : the dependent variable, for example the VIX at the close of day t
- ▶ S_{t_i} : the index level at the close of day t_i ; $t_i \leq t$ runs over the days up to and including t
- ▶ $r_{t_i} = S_{t_i}/S_{t_{i-1}} - 1$: the simple return of day t_i
- ▶ $R_{1,t}$: the trend component, built from past returns
- ▶ Σ_t : the volatility component, built from past squared returns
- ▶ $\beta_0, \beta_1, \beta_2$: coefficients estimated from the data

Case Study (4/14): The Two Components

- ▣ Trend component: a weighted average of past returns

$$R_{1,t} = \sum_{t_i \leq t} K_1(t - t_i) r_{t_i}$$

- ▶ $t - t_i$: the age of day t_i , in days; $K_1(t - t_i)$: the weight of that day
- ▶ $R_{1,t} < 0$ after a run of falls, $R_{1,t} > 0$ after a run of rises

- ▣ Volatility component: the square root of a weighted average of squared returns

$$\Sigma_t = \left(\sum_{t_i \leq t} K_2(t - t_i) r_{t_i}^2 \right)^{1/2}$$

- ▶ Σ_t is a weighted historical volatility, with weights K_2

- ▣ Interpreting the coefficients

- ▶ $\beta_1 < 0$: volatility rises after falls ($R_{1,t} < 0$), i.e. the leverage effect
- ▶ $\beta_2 \in (0, 1)$: high recent volatility raises today's volatility, i.e. volatility clustering, as in GARCH

Case Study (5/14): Weighting Kernels

- ▣ A **kernel** $K(\tau)$ gives the weight of a day that is τ days old; the weights are positive and sum to 1
- ▣ The paper uses a **time-shifted power law (TSPL)**, eq. (3.9)

$$K(\tau) = \frac{(\tau + \delta)^{-\alpha}}{Z_{\alpha, \delta}}, \quad \alpha > 1, \delta > 0, \tau = 0, 1, \dots, 999$$

- ▶ $Z_{\alpha, \delta}$: the normalising constant, the sum of $(\tau + \delta)^{-\alpha}$ over the last 1,000 days
- ▶ α : how fast the weights decay; a small α means long memory
- ▶ δ : the time shift; a small δ concentrates the weight on the last few days
- ▣ Resulting shape
 - ▶ large weight on the last few days (short memory)
 - ▶ a slowly decaying power-law tail (long memory, Fact 8)
- ▣ K_1 and K_2 each have their own parameters: (α_1, δ_1) and (α_2, δ_2)

Case Study (6/14): Estimation and Scores

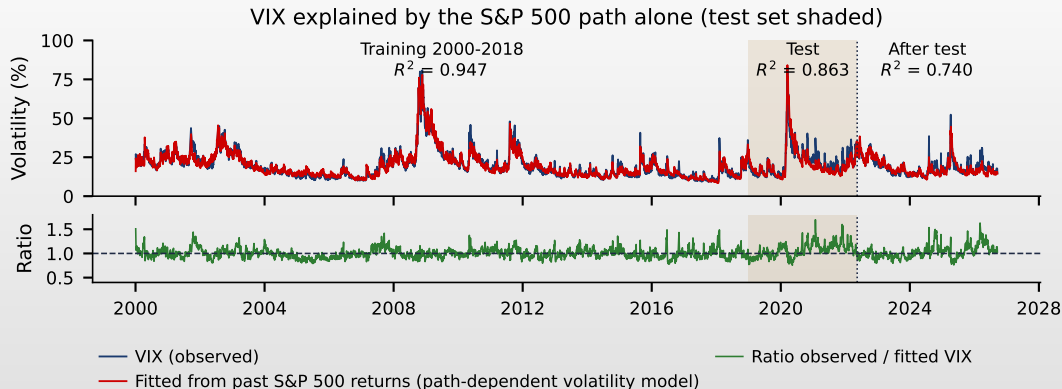
- ▣ Seven parameters: $\beta_0, \beta_1, \beta_2, \alpha_1, \delta_1, \alpha_2, \delta_2$
 - ▶ estimated by nonlinear least squares on the training set: minimise $\sum_t (\sigma_t - \hat{\sigma}_t)^2$
 - ▶ $\hat{\sigma}_t$: the model value for day t
 - ▶ estimation details: [▶ Appendix](#)
- ▣ Scores, computed separately on the training and the test set
 - ▶ RMSE (root mean squared error) = $\sqrt{\frac{1}{T} \sum_t (\sigma_t - \hat{\sigma}_t)^2}$: the typical error, in volatility units
 - ▶ $R^2 = 1 - \sum_t (\sigma_t - \hat{\sigma}_t)^2 / \sum_t (\sigma_t - \bar{\sigma})^2$: the share of the variation of σ_t explained by the model
 - ▶ $\bar{\sigma}$: the mean of σ_t ; $R^2 = 1$ is a perfect fit, $R^2 = 0$ is no better than the mean

Case Study (7/14): R^2 on the Training and Test Samples

Index	Implied volatility (paper, Table 2)		Realised volatility (paper, Table 4)	
	R^2 train	R^2 test	R^2 train	R^2 test
S&P 500 (VIX)	0.946	0.855	0.738	0.654
S&P 500 (VIX), our replication	0.947	0.863	—	—
S&P 500 (9-day VIX)	0.876	0.914	—	—
Euro Stoxx 50	0.929	0.913	0.672	0.682
Nikkei 225	0.890	0.800	0.563	0.504

- Past index returns alone explain 80–92% of the variance of implied volatility out of sample
 - ▶ FTSE 100 and DAX in the [Appendix](#)
- On our data (S&P 500 and VIX): RMSE 0.020 training and 0.034 test
 - ▶ versus 0.020 and 0.035 in Table 2 of the paper
- Realised volatility scores are lower
 - ▶ 5-minute realised volatility is a noisy target (Remark 3 in the paper)
 - ▶ this course uses no 5-minute data, so Table 4 of the paper is not replicated

Case Study (8/14): Observed and Fitted VIX

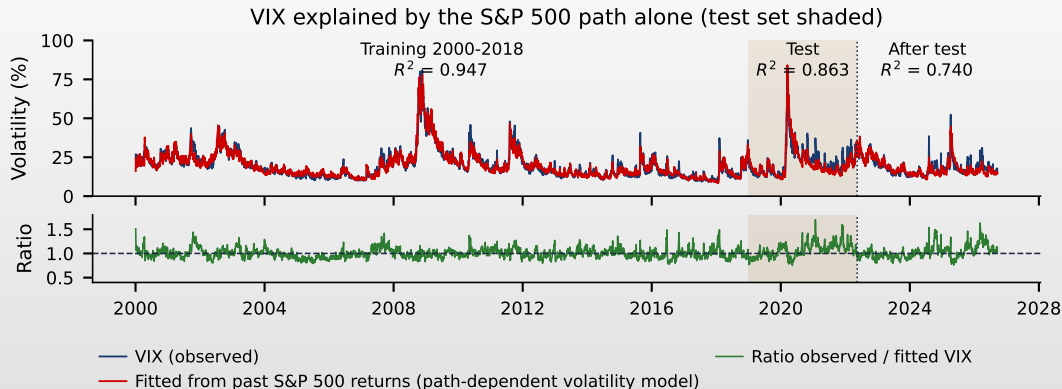


Case Study (9/14): Observed and Fitted VIX

- Fitted on 3 Jan 2000 – 31 Dec 2018 (4,779 days)
 - ▶ started from Table 3 of the paper
 - ▶ then parameters frozen
- The test set is shaded

 MFM_ch1_path_dependent_vol

Case Study (10/14): The VIX after the Test Set

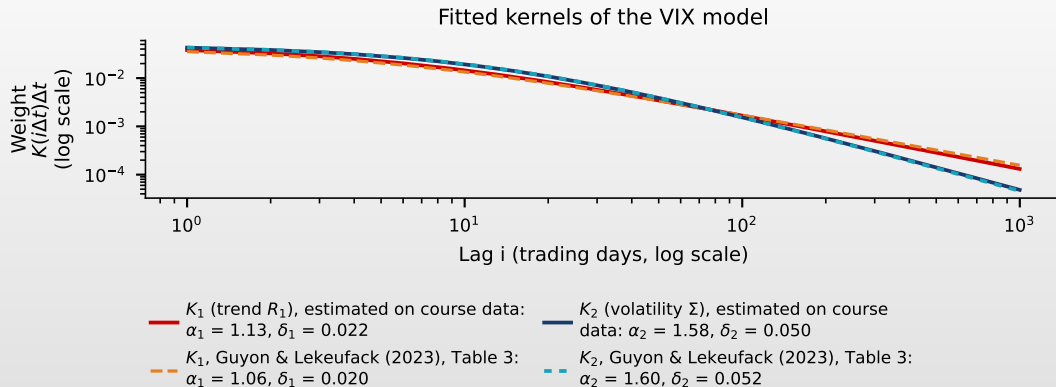


Case Study (11/14): The VIX after the Test Set

- After the test set, 16 May 2022 – 18 Sep 2026: $R^2 = 0.740$, RMSE 0.026
 - ▶ a lower RMSE, but the VIX varied less (standard deviation 5.0 versus 9.3 points)
 - ▶ so R^2 falls
- Ratio observed/fitted
 - ▶ mean 0.85 in April–May 2020
 - ▶ mean 1.16 from October 2020 to May 2022
 - ▶ 8 April 2025: VIX 52.3 versus 39.2 fitted

 MFM_ch1_path_dependent_vol

Case Study (12/14): Fitted Kernels



Case Study (13/14): Fitted Kernels

- Daily weights of K_2 , log-log scale
 - ▶ $\hat{\alpha}_2 = 1.58$, $\hat{\delta}_2 = 0.050$
 - ▶ versus 1.60, 0.052 in Table 3 of the paper
- The last 5 days carry 20% of the weight
 - ▶ the days beyond lag 250 still carry 10%
 - ▶ short and long memory in one kernel

 MFM_ch1_path_dependent_vol

Case Study (14/14): Limits and Extension

- ▣ Limits stated by the authors (Section 5 of the paper)
 - ▶ a small exogenous part remains
 - ▶ the ratio of observed to fitted VIX mean-reverts around 1
 - ▶ this ratio needs a stochastic factor
- ▣ Open extension (project seed)
 - ▶ does the same formula describe the realised volatility of Bitcoin or of BET-TR?

From Evidence to Model Requirements

Evidence (five markets)	Model requirement	Where in this course
quantile (Crow–Siddiqui) excess kurtosis > 0 in all five markets; Hill $\hat{\alpha} \approx 2.5\text{--}3.5$	heavy-tailed innovations, also conditionally	Ch. 5, Ch. 7 (extreme value theory)
ACF of $ r_t $ positive for months; $\hat{d} \approx 0.35\text{--}0.50$	persistent, time-varying volatility; long memory or breaks	Ch. 5, Ch. 9
$L(1) = -0.12$ (S&P 500, BET-TR); reversed for EUR/RON	asymmetric response to bad news where it exists	Ch. 5 (GJR-GARCH, EGARCH)
robust $\hat{\rho}_1$ economically small except EUR/RON	approximately uncorrelated returns; market microstructure matters	Ch. 2, Ch. 10
quantile skewness negative only for the S&P 500	asymmetry is market-specific	Ch. 7–8

- The random walk with i.i.d. increments from the Normal distribution (Bachelier–Black–Scholes world) violates Facts 2, 3 and 5–11
 - Fact 4 holds trivially
- ARCH-LM significance supports volatility clustering descriptively
 - with doubtful fourth moments it is not a formal proof

Open Question: Are Tail Indices Stable over Time? (1/2)

- The tail index α is usually treated as a constant
- Crypto-assets changed in structure
 - ▶ futures (2017), spot Bitcoin ETFs (January 2024), institutional holders
 - ▶ [Pele et al. \(2023\)](#): the tail factor separates cryptos from other asset classes
- Hypothesis proposed by an AI assistant
 - ▶ after the launch of the spot ETFs (January 2024), the entry of institutional investors has made the tails of Bitcoin's return distribution thinner
 - ▶ i.e. it has raised the tail index α

Open Question: Are Tail Indices Stable over Time? (2/2)

- ▣ Discovery loop in one pass
 - ▶ **literature**: ask for peer-reviewed work with DOIs, then resolve each DOI yourself
 - ▶ [Hoga \(2017\)](#): change-point tests for the tail index valid under β -mixing dependence
 - ▶ β -mixing: dependence that vanishes with the distance in time, e.g. in GARCH models
 - ▶ **hypotheses**: a change in tail shape, a fall in volatility only (scale does not move α), sampling noise
 - ▶ **data and code**: replicate the chapter's Hill estimates first (Bitcoin 4.2, S&P 500 3.1)
- ▣ Fixed in advance
 - ▶ $k = 2.5\%$ of n , the break date 11 January 2024
 - ▶ block-bootstrap SEs, a placebo asset

Mini-Case: Did Bitcoin's Tail Get Thinner after the ETFs? (1/2)

- Test: Hill estimate for $|r_t|$, $k = 2.5\%$ of n
 - ▶ $H_0 : \alpha_{\text{post}} = \alpha_{\text{pre}}$ against $H_1 : \alpha_{\text{post}} > \alpha_{\text{pre}}$
 - ▶ $\hat{\alpha}_{\text{pre}}, \hat{\alpha}_{\text{post}}$: the Hill estimates in the two subperiods; s : their standard errors, from a moving-block bootstrap (blocks of 20 days)
 - ▶ $z = (\hat{\alpha}_{\text{post}} - \hat{\alpha}_{\text{pre}}) / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$: the difference divided by its standard error
 - ▶ under H_0 , $z \approx N(0, 1)$; reject H_0 at 5% if $z > 1.645$
- Results
 - ▶ before (Sep 2014 – 10 Jan 2024, $k = 85$): $\hat{\alpha} = 4.16$ ($s = 0.52$)
 - ▶ after (11 Jan 2024 – Sep 2026, $k = 24$): $\hat{\alpha} = 4.31$ ($s = 0.91$)
 - ▶ $z = 0.15$
- Placebo, S&P 500 at the same date: $2.82 \rightarrow 2.86$, $z = 0.03$

Mini-Case: Did Bitcoin's Tail Get Thinner after the ETFs? (2/2)

- ▣ **Minimum detectable change**, one-sided 5% test with 80% power
 - ▶ $(1.645 + 0.842) \times 1.05 = 2.6$
 - ▶ 1.645: the one-sided 5% critical value; 0.842: the 80% quantile of $N(0, 1)$, for 80% power
 - ▶ $1.05 = \sqrt{0.52^2 + 0.91^2}$ is the SE of the difference
 - ▶ the Normal-distribution approximation is rough with 24 tail observations
 - ▶ the data cannot distinguish an unchanged tail from a large thinning of the tail
- ▣ A non-significant result does not show the absence of an effect
 - ▶ wait for more data or pool assets

Project Seed

- **Question:** did the tail index of Bitcoin and Ethereum change with institutionalisation (futures 2017, spot ETFs 2024)?
 - ▶ raw and GARCH-standardised returns (shape versus scale)
 - ▶ loss and gain tails separately
- **Replicate first:** the chapter's Hill table and the mini-case numbers
- **Pre-register**, committed before the results
 - ▶ H_0 , H_1 , the rule for k , break dates and bootstrap
- **Report:** the change-point test of [Hoga \(2017\)](#) as a Quantlet
 - ▶ estimates, both SEs, the minimum detectable change
- **Log:** prompts in AI_USE.md
 - ▶ every caught error (an i.i.d. SE, a date chosen from the data) in AI_ERRORS.md
 - ▶ reusable prompts (▶ [Appendix](#)) and checks (▶ [Appendix](#))

Key Takeaways and the Link to Chapter 2

Takeaways

- Returns are uncorrelated but not independent: their size is predictable
- Heavy tails are the rule: tail indices near 3
 - ▶ the fourth moment very likely does not exist
 - ▶ kurtosis-based inference fails
- Volatility is persistent (long memory or breaks)
 - ▶ in equity markets it reacts asymmetrically to losses
- Every test needs a reference distribution that the data may violate

Check yourself

- Why does the $Q(10)$ on Bitcoin returns have $p = 0.028$ but the robust $\tilde{Q}(10)$ $p = 0.29$?
- Why can the tail index be estimated with the Hill estimator when the kurtosis does not exist?
- **Chapter 2:** are returns predictable at all?

Appendix: Formula Sheet with Assumptions

Returns and annualisation

- ▣ $R_t = P_t/P_{t-1} - 1$, $r_t = \ln(1 + R_t)$, $r_t(k) = \sum_{j=0}^{k-1} r_{t-j}$,
 $R_{p,t} = \sum_i w_i R_{i,t}$
- ▣ $\text{CAGR} = (P_T/P_0)^{1/Y} - 1$
- ▣ $\sigma_{\text{ann}} = \sqrt{A} \hat{\sigma}$: uncorrelated returns, constant variance

Tests

- ▣ $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$: χ_2^2 under i.i.d. Normal
- ▣ $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k)$: χ_m^2 under i.i.d.
- ▣ $\tilde{Q}(m) = \sum_k \hat{\rho}_k^2 / \hat{\tau}_k$: $\mathbb{E}r^4 < \infty$, zero cross terms
- ▣ Hill $\text{SE}(\hat{\alpha}) \approx \hat{\alpha}/\sqrt{k}$: independent Pareto tail

Volatility

- ▣ Parkinson $\frac{1}{4 \ln 2} \overline{(h-l)^2}$, the bar a mean over days:
 driftless, no overnight jumps
- ▣ Mixture $K = 3 \mathbb{E}[\sigma^4] / \mathbb{E}[\sigma^2]^2$: σ_t independent of ε_t

Performance

- ▣ $SR = (\mu - r_f)/\sigma$ on simple returns
 ▶ $SE \approx 1/\sqrt{Y}$ for i.i.d. Normal returns
- ▣ $SoR = (\mu - r_f)/\sigma^-$
- ▣ $DD_t = P_t / \max_{s \leq t} P_s - 1$; Calmar = $\text{CAGR}/|MDD|$

Appendix: Prices Before Computers, the Stock Ticker



Stock ticker designed by Thomas Edison
(Smithsonian collection)

Photo: Jaclyn Nash, Smithsonian National Museum of
American History; public domain; Wikimedia Commons

- Before computers, trades reached brokers as prices printed on a paper **ticker tape**
- A daily close is one snapshot of this stream of prices
 - ▶ returns are built from such snapshots
- The choice of snapshot (close, fixing, a midnight snapshot) matters
 - ▶ as [Chapter 0](#) showed
- Prices at the level of individual trades: market microstructure, [Chapter 10](#)

Appendix: The People Behind the Stylised Facts (1/2)



Carl Friedrich Gauss (1777–1855)

Painting: C.A. Jensen (1840); public domain;
[Wikimedia Commons](#)

- The Normal distribution: the benchmark that returns violate



Louis Bachelier (1870–1946)

Photo: unknown author; public domain;
[Wikimedia Commons](#)

- Prices as a random walk, in his thesis [Bachelier \(1900\)](#)

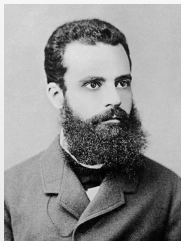


William S. Gosset, "Student" (1876–1937)

Photo: unknown author; public domain;
[Wikimedia Commons](#)

- The Student-*t* distribution [Student \(1908\)](#), today a model for heavy tails

Appendix: The People Behind the Stylised Facts (2/2)



Vilfredo Pareto (1848–1923)

Photo: unknown author (1870s); public domain; [Wikimedia Commons](#)



Benoît Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 France; [Wikimedia Commons](#)

- Pareto: **power laws** for large incomes, the ancestor of heavy-tail models
- Mandelbrot (1963): cotton prices
 - ▶ heavy tails, which the Normal distribution does not describe
- The exponent α of these power laws
 - ▶ is the tail index estimated with the Hill method

Appendix: Cont's Facts in Recent Studies

- Ratliff-Crain, Van Oort, Koehler & Tivnan (2025)
 - ▶ re-test all 11 facts on intraday returns of the Dow 30 stocks, October 2018 – March 2019
 - ▶ conclusive evidence for 8 facts, no support for the remaining 3
- Heavy tails beyond the Student- t : estimation of alpha-stable distributions, whose tail index is below 2 [Pele \(2014\)](#)
- Entropy-based measures of tail risk as an alternative to kurtosis [Pele, Lazar & Dufour \(2017\)](#)
- Crypto-assets: the same facts, with a tail factor that separates them from other asset classes [Pele et al. \(2023\)](#)

Appendix: Annual Variance under Autocorrelation (1/2)

- The variance of a sum of h daily returns depends on the autocorrelations

$$\text{Var}\left(\sum_{j=1}^h r_{t+j}\right) = h\sigma^2 \left[1 + 2 \sum_{j=1}^{h-1} (1 - j/h) \rho_j\right]$$

- ▶ h : the horizon in days; σ^2 : the daily variance; ρ_j : the autocorrelation at lag j
- ▶ assumption: weakly stationary returns (constant mean and variance, autocovariances depending only on the lag)
- ▶ without autocorrelation ($\rho_j = 0$) it is $h\sigma^2$, i.e. the $\sqrt{h}$ rule for volatility
- **Variance ratio** $VR(h) = \text{Var}(h\text{-day sum}) / (h\sigma^2)$
 - ▶ $VR(h) = 1$: no autocorrelation; $VR(h) > 1$: positive autocorrelation; $VR(h) < 1$: negative autocorrelation
 - ▶ volatility over horizon h is the $\sqrt{h}$ -rule value multiplied by $\sqrt{VR(h)}$

Appendix: Annual Variance under Autocorrelation (2/2)

- Autoregression of order 1, AR(1), with $\rho_j = \rho^j$: $\sqrt{VR(h)} \rightarrow \sqrt{(1+\rho)/(1-\rho)}$ as h grows
 - ▶ ρ : the lag-1 autocorrelation, $|\rho| < 1$
 - ▶ EUR/RON, $\rho_1 = 0.17$: factor 1.19, annual volatility 4.4% $\rightarrow$ 5.3%
 - ▶ S&P 500, $\rho_1 = -0.079$: factor 0.92
- Using the sample lag-1 autocorrelation as an AR(1) parameter is an assumption
 - ▶ check the whole ACF (Seminar A7)
 - ▶ variance-ratio tests: [Chapter 2](#)

Appendix: The Null Hypothesis of JB, and Tests for Time Series

- ▣ JB is a Lagrange Multiplier test [Jarque & Bera \(1980\)](#)
 - ▶ under i.i.d. returns from the Normal distribution, $\sqrt{T}(\hat{S}, \hat{K} - 3) \rightarrow N(0, \text{diag}(6, 24))$, hence χ^2_2
 - ▶ $\text{diag}(6, 24)$: a diagonal covariance matrix; $\hat{S}$ and $\hat{K}$ are asymptotically independent, with variances $6/T$ and $24/T$
 - ▶ JB is the sum of the two squared standardised statistics, hence χ^2 with 2 degrees of freedom
- ▣ GARCH-type dependence changes $\text{Var}(\hat{S})$ and $\text{Var}(\hat{K})$
 - ▶ a rejection mixes the marginal distribution and independence
- ▣ [Bai & Ng \(2005\)](#): tests of skewness, kurtosis and normality of the marginal distribution of a time series
 - ▶ studentised with HAC variances
 - ▶ their separate skewness and kurtosis tests (null $S = 0$ or $K = 3$ without normality) need $\mathbb{E}r^6 < \infty$ and $\mathbb{E}r^8 < \infty$

Appendix: The Robust Portmanteau Test and Its Assumptions

- $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k \sim \chi_m^2$ Lobato, Nankervis & Savin (2001) needs $\mathbb{E}r^4 < \infty$
 - ▶ $\sim$: "is distributed as", here asymptotically
 - ▶ and zero cross terms: $\mathbb{E}[x_t^2 x_{t-k} x_{t-l}] = 0$ for two different lags $k \neq l$
 - ▶ then the $\hat{\rho}_k$ at different lags are asymptotically uncorrelated and the sum is χ_m^2
- Dalla, Giraitis & Phillips (2022) keep only the significant cross terms in the covariance of the $\hat{\rho}_k$
 - ▶ none for the S&P 500, BET-TR and Bitcoin
- $\hat{\tau}_k$ is the variance under $\rho_k = 0$
 - ▶ to compare a non-zero ρ_1 across periods use the HAC SE of the slope of r_t on r_{t-1} (Seminar B11)
- ACF of $|r_t|$ and of r_t^2 : Normal limits need $\alpha > 4$ and $\alpha > 8$ Davis & Mikosch (1998)

Appendix: Hill Estimator, Asymptotic Distribution and Alternatives (1/2)

- ▣ Hill = maximum likelihood estimator of a Pareto tail above $u = X_{(k+1)}$, conditional on u (Seminar A11)
- ▣ If $k \rightarrow \infty$ and $k/n \rightarrow 0$, and the tail is Pareto up to a second-order term

$$\sqrt{k}(\hat{\alpha}_k - \alpha) \rightarrow N(0, \alpha^2)$$

- ▶ second-order term: a correction to the power law that vanishes as $x \rightarrow \infty$
- ▶ consequence: $SE(\hat{\alpha}_k) \approx \alpha/\sqrt{k}$, the formula in the main part
- ▶ if k grows too fast, a bias term appears: the tail is only approximately Pareto
- ▣ Dependent data (GARCH-type): $\hat{\alpha}$ stays consistent
 - ▶ but extremes cluster and the variance exceeds α^2/k [Hsing \(1991\)](#)

Appendix: Hill Estimator, Asymptotic Distribution and Alternatives (2/2)

- ▣ Choosing k : plot $\hat{\alpha}$ against $\log k$ ("alt-Hill" plot) Drees, Resnick & de Haan (2000)
 - ▶ the log scale stretches the region of small k , where the actual tail lies
- ▣ Regression alternative Gabaix & Ibragimov (2011)
 - ▶ log rank on log size, with rank shifted by $-\frac{1}{2}$: $\ln(i - \frac{1}{2}) = a - b \ln X_{(i)}$, $i = 1, \dots, k$
 - ▶ the slope $\hat{b}$ estimates α ; the shift by $\frac{1}{2}$ reduces the small-sample bias
 - ▶ $SE = \hat{\alpha} \sqrt{2/k}$

Appendix: Two Tails and the Tail of $|r_t|$

- $P(|X| > x) = P(X > x) + P(X < -x)$
 - ▶ with indices α_- (losses) and α_+ (gains), $|X|$ has index $\min(\alpha_-, \alpha_+)$
- At the same k the estimates are not comparable
 - ▶ the k largest $|r_t|$ come from both tails
 - ▶ common threshold u : the lower of the two one-sided thresholds at $k = 2.5\%$ of n

	u (%)	k for $ r_t $	$\hat{\alpha}, r_t $	$\hat{\alpha}_-, \text{losses}$	$\hat{\alpha}_+, \text{gains}$
S&P 500	2.17	508	2.90	2.85	2.96
BET-TR	1.79	153	2.54	2.20	3.06
Bitcoin	7.11	233	3.02	2.71	3.36
EUR/RON	0.54	295	2.28	2.63	2.04
Gold	1.98	513	3.11	2.97	3.28

- At a common u , the $|r_t|$ estimate lies between the two tails
 - ▶ Bitcoin's 4.2 at $k = 2.5\%$ reflects second-order bias
 - ▶ report both one-sided indices

Appendix: Estimating d , the Formulas (1/2)

- ▣ Fourier frequencies $\lambda_j = 2\pi j/T$, $j = 1, \dots, m$; the m lowest are used, here $m = T^{0.65}$
- ▣ The **periodogram**: the sample estimator of the spectral density

$$I(\lambda_j) = \frac{1}{2\pi T} \left| \sum_{t=1}^T x_t e^{-i\lambda_j t} \right|^2$$

- ▶ x_t : the demeaned series (here $|r_t|$ minus its mean); i : the imaginary unit
- ▶ $I(\lambda_j)$: how much of the variation in x_t comes from the cycle of frequency λ_j
- ▣ Log-periodogram regression [Geweke & Porter-Hudak \(1983\)](#): regress $\ln I(\lambda_j)$ on $-2 \ln(2 \sin(\lambda_j/2))$
 - ▶ the reason: $\ln f(\lambda) \approx \ln G - 2d \ln \lambda$, and $2 \sin(\lambda/2) \approx \lambda$ at low frequencies
 - ▶ the slope is $\hat{d}$, $SE = \pi/\sqrt{24m}$

Appendix: Estimating d , the Formulas (2/2)

- Local Whittle [Robinson \(1995\)](#): $\hat{d}$ minimises an objective built from the approximation $f(\lambda) \approx G\lambda^{-2d}$

$$\hat{d} = \arg \min_d \left[\ln \left(\frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^m \ln \lambda_j \right]$$

- ▶ $\arg \min_d$: the value of d that minimises the expression; the constant G has been eliminated analytically
 - ▶ $\sqrt{m}(\hat{d} - d) \rightarrow N(0, \frac{1}{4})$, so $SE = 1/(2\sqrt{m})$
- Local Whittle is robust to non-Gaussianity
 - ▶ a larger m lowers the SE
 - ▶ but lets short-run (GARCH) dynamics bias $\hat{d}$ (Seminar B12)

Appendix: Markets in Detail (1/2)

- ▣ **Crypto-assets:** an asset class with its own statistical profile [Härdle, Harvey & Reule \(2020\)](#)
 - ▶ tail, memory and moment factors classify cryptos as a separate asset class [Pele et al. \(2023\)](#)
 - ▶ the separation comes mainly through the tail factor
 - ▶ crypto modelling and the CRIX (CRypto IndeX): [Härdle, Chen & Overbeck \(2017\)](#); Chapter 16
- ▣ **EUR/RON:** BNR reference rate from 1 July 2005
 - ▶ the new leu (RON, 1 RON = 10,000 old lei)
 - ▶ long calm periods interrupted by sudden adjustments
 - ▶ EUR/RON market (interbank) quote series showed spurious negative autocorrelation (quote errors)

Appendix: Markets in Detail (2/2)

- ▣ **BET-TR:** the bank-asset tax was announced on 18 December 2018
 - ▶ adopted on 28 December 2018 as Government Emergency Ordinance (GEO) 114/2018
 - ▶ policy shocks produce single-day crashes without comparable rallies
- ▣ **Non-synchronous trading:** index constituents do not all trade at every moment
 - ▶ so the index reflects news with a delay
 - ▶ creating positive autocorrelation [Campbell, Lo & MacKinlay \(1997\)](#), ch. 3

Appendix: The Maximum Drawdown of a Brownian Motion

- ▣ Driftless Brownian motion of the log price with annual volatility σ over T years
 - ▶ here T is the horizon in years, not the number of days
- ▣ The maximum **log** drawdown $D = -\ln(1 + MDD) \geq 0$, the largest fall in the log price, has

$$\mathbb{E}[D] = \sqrt{\pi/2} \sigma \sqrt{T} \quad \text{Magdon-Ismail, Atiya, Pratap \& Abu-Mostafa (2004)}$$

- ▣ $MDD = e^{-D} - 1$; $\sqrt{\pi/2} \approx 1.25$
 - ▶ without drift the expected maximum drawdown grows in proportion to $\sqrt{T}$
 - ▶ with a positive drift $\mathbb{E}[D]$ grows only like $\ln T$
- ▣ Gold 1990–2026: $\sigma = 16.3\%$, $T = 36.7$
 - ▶ benchmark $\mathbb{E}[D] = 1.24$ versus observed 0.592 (−44.7%) (Seminar A2)
- ▣ Drawdowns and Calmar ratios from samples of different length are not comparable

Appendix: Sharpe Ratios beyond the Normal Distribution

- i.i.d., any distribution with $\mathbb{E}r^4 < \infty$ [Opdyke \(2007\)](#)
 - ▶ $\text{Var}(\widehat{SR}_d) \approx (1 - S SR_d + \frac{K-1}{4} SR_d^2) / T$
 - ▶ S, K : the skewness and kurtosis of daily returns; for the Normal distribution ($S = 0, K = 3$) this gives Lo's formula
 - ▶ negative skewness and high kurtosis raise the variance of $\widehat{SR}_d$ when $SR_d > 0$
 - ▶ here the SEs change by at most 0.02
- The Deflated Sharpe Ratio corrects for non-normality and multiple testing ([Chapter 13](#))

	CAGR	Sharpe	Sortino	Calmar
S&P 500	8.7%	0.55	0.78	0.15
BET-TR	21.2%	1.34	1.87	0.68
Bitcoin	53.9%	0.98	1.45	0.65
EUR/RON	1.8%	0.43	0.65	0.12
Gold	6.7%	0.48	0.69	0.15

Appendix: Case Study, Estimation Details (1/2)

- Targets (Sections 3.4–3.5 of the paper)
 - ▶ the 30-day implied volatility index of each index and the 9-day VIX
 - ▶ realised volatility of day $t + 1$
 - ▶ volatility in decimals ($0.03 = 3$ volatility points)
- Kernels, eq. (3.9) in the paper: $K(\tau) = Z_{\alpha, \delta}^{-1}(\tau + \delta)^{-\alpha}$
 - ▶ normalised so that $\sum_i K(i\Delta t)\Delta t = 1$ on the 1,000-day window
 - ▶ i : the age of the day, in days; $\Delta t = 1/252$: one day, in years
- Estimation: $(\beta_0, \beta_1, \beta_2, \alpha_1, \delta_1, \alpha_2, \delta_2)$ by nonlinear least squares on the training set
 - ▶ initialised using a lasso regression on features built with exponential kernels

Appendix: Case Study, Estimation Details (2/2)

- Our replication, started from the values in Table 3 of the paper (in parentheses)
 - ▶ $\hat{\beta}_0 = 0.052$ (0.057), $\hat{\beta}_1 = -0.090$ (-0.095)
 - ▶ $\hat{\beta}_2 = 0.868$ (0.82), $\hat{\alpha}_1 = 1.13$ (1.06)
 - ▶ the parameters differ slightly, but R^2 and RMSE are practically the same
- The 9-day VIX exists only from 2011 (shorter training set)
- The restricted models ($\beta_1 = 0$, $\beta_2 = 0$) are Seminar C3

Appendix: Case Study, R^2 for All Indices (1/2)

Index	Implied volatility (paper, Table 2)		Realised volatility (paper, Table 4)	
	R^2 train	R^2 test	R^2 train	R^2 test
S&P 500 (VIX)	0.946	0.855	0.738	0.654
S&P 500 (9-day VIX)	0.876	0.914	—	—
Euro Stoxx 50	0.929	0.913	0.672	0.682
FTSE 100	0.925	0.870	0.650	0.617
DAX	0.934	0.918	0.722	0.557
Nikkei 225	0.890	0.800	0.563	0.504

- VIX parameters: $\beta_0 = 0.057$, $\beta_1 = -0.095$, $\beta_2 = 0.82$
- Across all targets $\alpha_2 \in [1.2, 2.3]$ (Tables 3 and 5 of the paper)
 - long memory near criticality (Section 3.2 of the paper)

Appendix: Case Study, R^2 for All Indices (2/2)

- Dropping Σ_t hurts the most
- The model beats two ARCH benchmarks and four published path-dependent models
 - ▶ Figures 3.8–3.9 and Appendix C of the paper
- Its Markovian 4-factor version, driven by Brownian motion (Section 4.6 and Table 9 of the paper)
 - ▶ gives estimated values of about 0.12–0.17 of the Hurst exponent H for daily realised volatility, as in data
 - ▶ $H < 0.5$: a path “rougher” than Brownian motion
 - ▶ a model without rough volatility reproduces the small values of H estimated from daily data

Appendix: The Discovery Loop, Reusable Prompts (1/2)

□ Literature

- ▶ Find peer-reviewed work since 2015 on time variation or structural breaks in tail indices of asset returns, with tests valid under GARCH-type dependence.
- ▶ For each: DOI, estimator, test, data.
- ▶ Flag entries you are unsure of.

□ Hypothesis

- ▶ State competing hypotheses for a change in Bitcoin's tail after the 2024 spot ETFs.
- ▶ What does each imply for the Hill estimate of the tail index of raw and of GARCH-standardised returns?

Appendix: The Discovery Loop, Reusable Prompts (2/2)

□ Data and code

- ▶ Use daily Bitcoin closing prices in USD; log returns on its own calendar;
- ▶ Hill estimate of the tail index for $|r|$, with $k = 2.5\%$ of n , before and after 11 January 2024; moving-block bootstrap standard errors.

□ Robustness

- ▶ Write a pre-registered plan: k from 1% to 5% of n , loss and gain tails, GARCH-standardised residuals, block lengths 10 to 50 days, a placebo date and a placebo asset.

□ Critique

- ▶ Referee this design.
- ▶ With 24 tail observations after the break, what is the power against a change of 1 in the tail index?

Appendix: Mandatory Checks for the Researcher (1/2)

- **Wrong return or price series:** reproduce the chapter's Hill table before anything else
- **Choice of k after seeing the result**
 - ▶ fix the rule ($k = 2.5\%$ of n) in advance
 - ▶ show the Hill plot for 1–5%
- **i.i.d. standard errors:** $\hat{\alpha}/\sqrt{k}$ ignores volatility clustering
 - ▶ compare with a block bootstrap
 - ▶ for the S&P 500 before 2024 the bootstrap SE is 2.2 times larger

Appendix: Mandatory Checks for the Researcher (2/2)

- **Break date chosen from the data**
 - ▶ use a date fixed by the event
 - ▶ or a sup-type change-point test with its own critical values [Hoga \(2017\)](#)
- **Bad ticks create fake tails**
 - ▶ list the ten largest $|r_t|$ and match each to a dated event
- **Low power:** report the minimum detectable change
 - ▶ 2.8 SEs of the difference for a two-sided 5% test with 80% power
 - ▶ 2.5 SEs for a one-sided test

Appendix: Practical Checklist for Empirical Work (1/2)

□ **Know your data**

- ▶ price vs total-return index
- ▶ adjusted vs raw prices
- ▶ fixing vs traded price
- ▶ official reference rates where possible (EUR/RON)

□ **Clean before you model**

- ▶ non-trading days, holidays, stale quotes, obvious errors
- ▶ match the largest moves to real events

□ **Use log returns** for single-asset dynamics, simple returns for portfolios

Appendix: Practical Checklist for Empirical Work (2/2)

- ▣ **Report robust statistics:** quantiles, the number of tail days and the tail index $\hat{\alpha}$ next to kurtosis
- ▣ **Annualise carefully**
 - ▶ 252 vs 365 periods
 - ▶ the $\sqrt{A}$ rule needs uncorrelated returns with constant variance
- ▣ **Reproducibility:** code and data on GitHub, as Quantlets

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Tukey's g- and h- transformations
- ▣  Applied Time Series Analysis with Python

Code, data and essential reading

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_01
- ▣ Essential reading: Cont (2001); Franke, Härdle & Hafner (2019); Campbell, Lo & MacKinlay (1997), ch. 1–2

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